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10 Additional tools

At times, it is useful to resort to some of the more advanced tools Riemannian geometry has to offer. We discuss some of these here in relation to optimization. The background on differential geometry given in Chapters 3 and 5 is often sufficient. We omit classical proofs, pointing to standard texts instead.

We start with the notion of Riemannian distance, which allows us to turn a (connected) Riemannian manifold into a metric space. It turns out that the associated metric space topology coincides with the manifold topology, and that shortest paths between pairs of points are geodesics. From there, we discuss the Riemannian exponential map: this is a retraction whose curves are geodesics. Then, we also give a formal and not-so-standard treatment of the inverse of the exponential map and, more generally, of the inverse of retractions.

Moving on, parallel transports allow us to move tangent vectors around, from tangent space to tangent space, isometrically. Combined with the exponential map, this tool makes it possible to define a notion of Lipschitz continuity for the gradient and the Hessian of a cost function on a Riemannian manifold. This leads to a sharp understanding of the regularity assumptions we made in Chapters 4 and 6 to control the worst-case behavior of optimization algorithms.

We follow up with the (also not-so-standard) notion of transporter—a poor man's version of parallel transport. This is useful to design certain algorithms. It also affords us a practical notion of finite difference approximation for the Hessian, which makes it possible to use second-order optimization algorithms without computing the Hessian.

In closing, we discuss covariant differentiation of tensor fields of any order.

As a notable omission, we do not discuss curvature at all: see for example [Lee18, Ch. 1, 7] for an introduction.

10.1 Distance, geodesics and completeness

Two points of $\mathcal{M}$ belong to the same *connected component* if there exists a continuous curve on $\mathcal{M}$ joining them. We say $\mathcal{M}$ is *connected* if it has a single connected component. A manifold has finitely many, or countably infinitely many connected components, owing to second-countability of the atlas topology (see Section 8.2).

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Definition 10.1. *A* distance *on a set $\mathcal{M}$ is a function* $\operatorname{dist}\colon \mathcal{M} \times \mathcal{M} \to \mathbb{R}$ *such that, for all $x, y, z \in \mathcal{M}$,*

1. $\operatorname{dist}(x, y) = \operatorname{dist}(y, x)$;
2. $\operatorname{dist}(x, y) \geq 0$, *and* $\operatorname{dist}(x, y) = 0$ *if and only* $x = y$; *and*
3. $\operatorname{dist}(x, z) \leq \operatorname{dist}(x, y) + \operatorname{dist}(y, z)$.

Equipped with a distance, $\mathcal{M}$ is a metric space.

A natural topology on a metric space is the *metric topology*, defined such that the functions $x \mapsto \operatorname{dist}(x, y)$ are continuous. Specifically, a subset $\mathcal{U} \subseteq \mathcal{M}$ is open if and only if, for every $x \in \mathcal{U}$, there exists a radius $r > 0$ such that the ball $\{y \in \mathcal{M} : \operatorname{dist}(x, y) < r\}$ is included in $\mathcal{U}$.

In this section, we first state without proof that if $\mathcal{M}$ is a Riemannian manifold then its Riemannian metric induces a distance on (the connected components of) $\mathcal{M}$, and that the topology of $\mathcal{M}$ as a manifold is equivalent to the topology of $\mathcal{M}$ as a metric space equipped with that distance. Intuitively, $\operatorname{dist}(x, y)$ is the length of the shortest "reasonable" curve on $\mathcal{M}$ joining x and y, or the infimum over the lengths of such curves. To make this precise, we first need to discuss various types of curves on manifolds [Lee18, pp33–34].

Definition 10.2. *A* curve segment *on a manifold $\mathcal{M}$ is a continuous map $c\colon [a, b] \to \mathcal{M}$, where $a \leq b$ are real. A curve segment $c\colon [a, b] \to \mathcal{M}$ is:*

- smooth *if c can be extended to a smooth map $\tilde{c}\colon I \to \mathcal{M}$ on a neighborhood I of $[a, b]$, in which case $c'(a)$ and $c'(b)$ denote $\tilde{c}'(a)$ and $\tilde{c}'(b)$, respectively;*
- regular *if it is smooth and $c'(t) \neq 0$ for all $t \in [a, b]$;*
- piecewise smooth *(resp.,* piecewise regular*) if there exists a finite set of times $a = t_0 < t_1 < \cdots < t_{k-1} < t_k = b$ such that the restrictions $c|_{[t_{i-1}, t_i]}$ are smooth (resp., regular) curve segments for $i = 1, \ldots, k$.*

In particular, piecewise regular curves are piecewise smooth. We say a curve segment $c\colon [a, b] \to \mathcal{M}$ connects *x to y if $c(a) = x$ and $c(b) = y$.*

Let $\mathcal{M}$ be a Riemannian manifold. Given a piecewise smooth curve segment $c\colon [a, b] \to \mathcal{M}$, we define the *length* of c as the integral of its *speed* $\|c'(t)\|_{c(t)}$:

$$L(c) = \int_a^b \|c'(t)\|_{c(t)} \, \mathrm{d}t. \tag{10.1}$$

While $c'(t_i)$ may be undefined, the speed of each curve segment $c|_{[t_{i-1}, t_i]}$ is smooth: the integral is computed by summing over these intervals.

The notion of length of a curve leads to a natural notion of distance on $\mathcal{M}$, called the *Riemannian distance*:

$$\operatorname{dist}(x, y) = \inf_c L(c), \tag{10.2}$$

where the infimum is taken over all piecewise regular curve segments on $\mathcal{M}$ which

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connect x to y. It is equivalent to take the infimum over all piecewise smooth curve segments. We have the following important result [Lee18, Thm. 2.55].

Theorem 10.3. *If $\mathcal{M}$ is connected (meaning each pair of points is connected by a curve segment), equation (10.2) defines a distance. Equipped with this distance, $\mathcal{M}$ is a metric space whose metric topology coincides with its atlas topology.*

If $\mathcal{M}$ is not connected, we may consider this result on the connected components of $\mathcal{M}$ separately. Sometimes, it helps to extend the definition to accept $\operatorname{dist}(x, y) = \infty$ when x, y belong to distinct connected components.

If the infimum in (10.2) is attained[1] for some curve segment c, we call c a *minimizing curve*. Remarkably, up to parameterization, these are geodesics (Definition 5.38) [Lee18, Thm. 6.4]. In other words, two competing generalizations of the notion of straight line from linear spaces to manifolds turn out to be equivalent: one based on shortest paths, one based on zero acceleration.

Theorem 10.4. *Every minimizing curve admits a constant-speed parameterization such that it is a geodesic, called a* minimizing geodesic*.*

This theorem admits a partial converse [Lee18, Thm. 6.15]. It could not have a full converse since, for example, two nearby points on a sphere can be connected through both a short and a long geodesic.

Theorem 10.5. *Every geodesic γ on $\mathcal{M}$ is* locally minimizing*, that is, every t in the domain of γ has a neighborhood I in the domain of γ such that, if $a, b \in I$ satisfy $a < b$, then the restriction $\gamma|_{[a,b]}$ is a minimizing curve.*

Equipped with a distance, we define a first notion of *completeness*. Recall that a sequence $x_0, x_1, x_2, \ldots$ is *Cauchy* if for every $\varepsilon > 0$ there exists an integer k such that, for all $m, n > k$, $\operatorname{dist}(x_m, x_n) < \varepsilon$.

Definition 10.6. *A connected Riemannian manifold is* metrically complete *if it is complete as a metric space equipped with the Riemannian distance, that is, if every Cauchy sequence on the manifold converges on the manifold.*

There exists another useful notion of completeness for manifolds.

Definition 10.7. *A Riemannian manifold is* geodesically complete *if every geodesic can be extended to a geodesic defined on the whole real line.*

The following theorem is an important classical result: see [Lee18, Thm. 6.19, Pb. 6-14] for a proof. It justifies omitting to specify whether we mean metric or geodesic completeness. The last part of the statement is the *Heine–Borel property*: recall Definition 8.26 for compact sets; a set S is *bounded* if $\sup_{x,y \in S} \operatorname{dist}(x, y)$ is finite.

[1] This is not always the case: think of $\mathcal{M} = \mathbb{R}^2 \backslash \{0\}$ as a Riemannian submanifold of $\mathbb{R}^2$, and connect x and $-x$.

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Theorem 10.8 (Hopf–Rinow)**.** *A connected Riemannian manifold $\mathcal{M}$ is metrically complete if and only if it is geodesically complete. Additionally, $\mathcal{M}$ is complete (in either sense) if and only if its compact subsets are exactly its closed and bounded subsets.*

For disconnected manifolds, *complete* refers to geodesic completeness, which is equivalent to metric completeness of each connected component. For example, the orthogonal group $\mathrm{O}(n)$, which has two connected components, is complete in this sense.

On a complete manifold, two points in the same connected component can always be connected by a (not necessarily unique) geodesic which attains the infimum in (10.2) [Lee18, Cor. 6.21].

Theorem 10.9. *If $\mathcal{M}$ is complete, then any two points x, y in the same connected component are connected by a minimizing geodesic segment $c\colon [0,1] \to \mathcal{M}$ such that $c(0) = x$, $c(1) = y$ and $\operatorname{dist}(x,y) = L(c)$.*

The converse of Theorem 10.9 does not hold: consider $\mathcal{M} = (0,1)$ as a Riemannian submanifold of $\mathbb{R}$.

Example 10.10. *Compact Riemannian manifolds are complete.*

Example 10.11. *A finite-dimensional Euclidean space $\mathcal{E}$ is connected and complete. The unique minimizing geodesic from x to y is the line segment $t \mapsto (1-t)x + ty$ on $[0,1]$, and the Riemannian distance $\operatorname{dist}(x,y)$ is equal to the Euclidean distance $\|x - y\|$.*

Exercise 10.12. *Let $\mathcal{M}$ be a Riemanian submanifold of $\mathcal{N}$. Show that if $\mathcal{N}$ is complete and if $\mathcal{M}$ is a closed subset of $\mathcal{N}$, then $\mathcal{M}$ is complete. In particular, Riemannian submanifolds of Euclidean spaces which are closed are complete.*

Exercise 10.13. *Show that the length of a piecewise smooth curve segment $c\colon [a,b] \to \mathcal{M}$ is independent of parameterization, in that $L(c \circ h) = L(c)$ for any monotone, piecewise regular $h\colon [0,1] \to [a,b]$ such that $h(0) = a$ and $h(1) = b$. Further show that if c is a piecewise regular curve segment then h can be chosen such that $c \circ h$ is piecewise regular with speed (whenever it is defined) equal to the constant $L(c)$.*

Exercise 10.14. *Show that the distance on a Riemannian product manifold $\mathcal{M} = \mathcal{M}_1 \times \cdots \times \mathcal{M}_n$ is given by*

$$\operatorname{dist}(x,y) = \sqrt{\sum_{i=1}^{n} \operatorname{dist}(x_i, y_i)^2},$$

where dist *denotes Riemannian distance on $\mathcal{M}$ and $\mathcal{M}_1, \ldots, \mathcal{M}_n$ alike, as indicated by context. Hint: the equality can be established through a pair of matching inequalities. For one side, Jensen's inequality may be helpful. For the other side, it may be helpful to use the results of Exercise 10.13.*

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Exercise 10.15. *Let $\mathcal{M} = \overline{\mathcal{M}}/\sim$ be a Riemannian quotient manifold with quotient map $\pi\colon \overline{\mathcal{M}} \to \mathcal{M}$, as in Section 9.7. Let* $\mathrm{dist}_{\mathcal{M}}$ *and* $\mathrm{dist}_{\overline{\mathcal{M}}}$ *denote the Riemannian distances on $\mathcal{M}$ and $\overline{\mathcal{M}}$, respectively. Show that for all $[x], [y] \in \mathcal{M}$ the distances satisfy*

$$\mathrm{dist}_{\mathcal{M}}([x], [y]) \leq \inf_{x' \sim x, y' \sim y} \mathrm{dist}_{\overline{\mathcal{M}}}(x', y').$$

Moreover, show that if $\overline{\mathcal{M}}$ is connected and complete then the inequality holds with equality. Hint: for the second part, use Proposition 9.61.

10.2 Exponential and logarithmic maps

Using standard tools from the study of ordinary differential equations, one can show that on a Riemannian manifold, for every $(x, v) \in \mathrm{T}\mathcal{M}$, there exists a unique *maximal* geodesic [Lee18, Cor. 4.28]

$$\gamma_v\colon I \to \mathcal{M}, \qquad \text{with} \qquad \gamma_v(0) = x \qquad \text{and} \qquad \gamma_v'(0) = v.$$

Here, *maximal* refers to the fact that the interval I is as large as possible (this is *not* in contrast to the notion of minimizing geodesic we just defined in the previous section). We use these geodesics to define a special map.

Definition 10.16. *Consider the following subset of the tangent bundle:*

$$\mathcal{O} = \{(x, v) \in \mathrm{T}\mathcal{M} : \gamma_v \text{ is defined on an interval containing } [0, 1]\}.$$

The exponential map $\mathrm{Exp}\colon \mathcal{O} \to \mathcal{M}$ *is defined by*

$$\mathrm{Exp}(x, v) = \mathrm{Exp}_x(v) = \gamma_v(1).$$

The restriction Exp_x *is defined on $\mathcal{O}_x = \{v \in \mathrm{T}_x\mathcal{M} : (x, v) \in \mathcal{O}\}$.*

For example, in a Euclidean space, $\mathrm{Exp}_x(v) = x + v$. By Definition 10.7, a manifold $\mathcal{M}$ is (geodesically) complete exactly if the domain of the exponential map is the whole tangent bundle $\mathrm{T}\mathcal{M}$.

Given $t \in \mathbb{R}$, it holds that $\gamma_{tv}(1) = \gamma_v(t)$ whenever either is defined. This allows us to express the exponential map as

$$\mathrm{Exp}_x(tv) = \gamma_v(t), \tag{10.3}$$

which is often more practical. In particular, the domain of Exp_x is *star-shaped* around the origin in $\mathrm{T}_x\mathcal{M}$, that is,

$$v \in \mathcal{O}_x \implies tv \in \mathcal{O}_x \text{ for all } t \in [0, 1].$$

Conveniently, Exp is smooth [Lee18, Lem. 5.18, Prop. 5.19].

Proposition 10.17. *The exponential map is smooth on its domain $\mathcal{O}$, which is open in $\mathrm{T}\mathcal{M}$.*

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The domain $\mathcal{O}$ contains all tangent space origins. We say that $\mathcal{O}$ is a neighborhood of the *zero section of the tangent bundle*:

$$\{(x, 0) \in \mathrm{T}\mathcal{M} : x \in \mathcal{M}\} \subset \mathcal{O}. \tag{10.4}$$

The exponential map is a retraction on its domain. More precisely:

Proposition 10.18. *The exponential map is a second-order retraction, with a possibly restricted domain* $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$.

Proof. Proposition 10.17 claims smoothness of Exp. By definition, for all (x, v) in $\mathrm{T}\mathcal{M}$ the curve $c(t) = \mathrm{Exp}_x(tv) = \gamma_v(t)$ satisfies $c(0) = x$ and $c'(0) = v$ so that Exp is a retraction. Finally, it is clear that this retraction is second order (Definition 5.42) since $\gamma_v''(t)$ is zero for all t, hence in particular $c''(0) = 0$. □

Given a point x and a (sufficiently short) tangent vector v, the exponential map produces a new point $y = \mathrm{Exp}_x(v)$. One may reasonably wonder whether, given the two points x, y, one can recover the tangent vector v. In what follows, we aim to understand to what extent the exponential map can be (smoothly) inverted.

A first observation, rooted in the inverse function theorem for manifolds (see Theorem 4.16), is that any retraction R at a point x is locally a diffeomorphism around the origin in the tangent space at x, because R_x is smooth and $\mathrm{DR}_x(0)$ is the identity (hence invertible). This applies in particular to the exponential map. For the latter, the *injectivity radius* quantifies how large the local domains can be.

When we say a map F is a diffeomorphism on an open domain U, we mean ⋆
that $F(U)$ is open and F is a diffeomorphism from U to $F(U)$.

Definition 10.19. *The* injectivity radius *of a Riemannian manifold* $\mathcal{M}$ *at a point* x*, denoted by* $\mathrm{inj}(x)$*, is the supremum over radii* $r > 0$ *such that* Exp_x *is defined and is a diffeomorphism on the open ball*

$$B(x, r) = \{v \in \mathrm{T}_x\mathcal{M} : \|v\|_x < r\}.$$

By the inverse function theorem, $\mathrm{inj}(x) > 0$.

Consider the ball $U = B(x, \mathrm{inj}(x))$ in the tangent space at x. Its image $\mathcal{U} = \mathrm{Exp}_x(U)$ is a neighborhood of x in $\mathcal{M}$. By definition, $\mathrm{Exp}_x \colon U \to \mathcal{U}$ is a diffeomorphism: it has a well-defined smooth inverse $\mathrm{Exp}_x^{-1} \colon \mathcal{U} \to U$. With these choices of domains, $v = \mathrm{Exp}_x^{-1}(y)$ is the unique shortest tangent vector at x such that $\mathrm{Exp}_x(v) = y$. Indeed, if there existed another vector $u \in \mathrm{T}_x\mathcal{M}$ such that $\mathrm{Exp}_x(u) = y$ and $\|u\|_x \leq \|v\|_x$, then u would be included in U, which would contradict invertibility. This motivates the following definition.[2]

[2] Note that Log_x may be discontinuous (think of x and y close to each other on a sphere from which we remove the midpoint $(x + y)/\|x + y\|$). Nevertheless, see below for smoothness properties on restricted domains (Corollary 10.25 in particular), and see also [Lee18, §10] for a related discussion of *cut locus* and *conjugate points*.

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Definition 10.20. *For $x \in \mathcal{M}$, let* Log_x *denote the* logarithmic map at x,

$$\mathrm{Log}_x(y) = \operatorname*{arg\,min}_{v \in \mathcal{O}_x} \|v\|_x \text{ subject to } \mathrm{Exp}_x(v) = y, \tag{10.5}$$

with domain such that this is uniquely defined.

For example, in a Euclidean space, $\mathrm{Log}_x(y) = y - x$ for all x, y. In particular, with domains $U = B(x, \mathrm{inj}(x))$ and $\mathcal{U} = \mathrm{Exp}_x(U)$ as above, the inverse of $\mathrm{Exp}_x \colon U \to \mathcal{U}$ is the (possibly restricted) map $\mathrm{Log}_x \colon \mathcal{U} \to U$, which is then a diffeomorphism. With different domain restrictions however, the inverse of Exp_x may be different from Log_x. This is illustrated in the following example.

Example 10.21. *On the sphere S^{n-1}, the exponential map is (Example 5.37)*

$$v \mapsto \mathrm{Exp}_x(v) = \cos(\|v\|)x + \frac{\sin(\|v\|)}{\|v\|}v.$$

This is smooth over the whole tangent bundle, with the usual smooth extension $\sin(t)/t = 1$ at $t = 0$. Given $x, y \in \mathrm{S}^{n-1}$, we seek an expression for $\mathrm{Exp}_x^{-1}(y)$. Since $x^\top x = 1$ and $x^\top v = 0$, considering $y = \mathrm{Exp}_x(v)$ as above, we deduce that $x^\top y = \cos(\|v\|)$. Thus, the following vector (which is readily computed given x and y) is parallel to v:

$$u \triangleq y - (x^\top y)x = \mathrm{Proj}_x(y) = \frac{\sin(\|v\|)}{\|v\|}v.$$

It has norm $|\sin(\|v\|)|$ and is parallel to v. Let us exclude the case $u = 0$ which is easily treated separately. Then, dividing u by its norm yields:

$$\frac{u}{\|u\|} = \mathrm{sign}(\sin(\|v\|))\frac{v}{\|v\|}.$$

If we restrict the domain of Exp_x to contain exactly those tangent vectors v whose norm is strictly less than π, then $\mathrm{sign}(\sin(\|v\|)) = 1$. Furthermore, the equation $x^\top y = \cos(\|v\|)$ then admits the unique solution $\|v\| = \arccos(x^\top y)$, where $\arccos \colon [-1, 1] \to [0, \pi]$ is the principal inverse of $\cos$. Overall, this yields the following expression:

$$y \mapsto \mathrm{Exp}_x^{-1}(y) = \arccos(x^\top y)\frac{u}{\|u\|}, \tag{10.6}$$

smooth over $\mathrm{S}^{n-1} \backslash \{-x\}$. Since the chosen domain for Exp_x is $B(x, \pi)$, the inverse is the logarithm: $\mathrm{Exp}_x^{-1} = \mathrm{Log}_x$. Also, $\mathrm{dist}(x, y) = \arccos(x^\top y)$.

The crucial point is that, in deriving this expression, we made the (somewhat arbitrary) choice of defining the domain of Exp_x in a specific way. This leads to a particular formula for the inverse, and a particular domain for Exp_x^{-1}. If we choose the domain of Exp_x differently, we may very well obtain a different formula for Exp_x^{-1} (not equal to Log_x) and a different domain for it as well. For example, on the circle S^1, we could decide that if y is ahead of x (counterclockwise) by an angle less than $\pi/2$, then $\mathrm{Exp}_x^{-1}(y)$ returns a vector of length

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less than $\pi/2$, and otherwise it returns a vector of length less than $3\pi/2$, pointing in the clockwise direction. In short, domains matter.

Before moving on to broader smoothness concerns, we quote useful relations between Exp, Log and dist [Lee18, Prop. 6.11].

Proposition 10.22. *If $\|v\|_x < \mathrm{inj}(x)$, the geodesic $c(t) = \mathrm{Exp}_x(tv)$ on the interval $[0, 1]$ is the minimizing curve connecting x to $y = \mathrm{Exp}_x(v)$, unique up to parameterization. In particular, $\mathrm{dist}(x, y) = \|v\|_x$, and $\mathrm{Log}_x(y) = v$.*

So far, we have fixed the point x, allowing us to claim that, on some domains, the map $y \mapsto \mathrm{Exp}_x^{-1}(y)$ is smooth in y. In order to discuss smoothness jointly in x and y, we need more work. We start with a general discussion valid for all retractions, and specialize to the exponential map later on.

Proposition 10.23. *Let $\mathcal{M}$ be a manifold with retraction R defined on a neighborhood $\mathcal{O}$ of the zero section of $\mathrm{T}\mathcal{M}$. Consider the following map:*

$$E\colon \mathcal{O} \to \mathcal{M} \times \mathcal{M}\colon (x, v) \mapsto E(x, v) = (x, \mathrm{R}_x(v)). \tag{10.7}$$

If $\mathcal{T} \subseteq \mathcal{O}$ is open in $\mathrm{T}\mathcal{M}$ such that, for all x, R_x is a diffeomorphism on $\mathcal{T}_x = \{v \in \mathrm{T}_x\mathcal{M} : (x, v) \in \mathcal{T}\}$, then $\mathcal{V} = E(\mathcal{T})$ is open in $\mathcal{M} \times \mathcal{M}$ and $E\colon \mathcal{T} \to \mathcal{V}$ is a diffeomorphism.

Proof. First, to see that $E\colon \mathcal{T} \to \mathcal{V}$ is invertible, consider any pairs $(x, v), (y, w) \in \mathcal{T}$ such that $E(x, v) = E(y, w)$. In other words, we have $(x, \mathrm{R}_x(v)) = (y, \mathrm{R}_y(w))$, so that $x = y$ and $v, w \in \mathcal{T}_x$. By assumption, R_x is injective on $\mathcal{T}_x$, hence we deduce from $\mathrm{R}_x(v) = \mathrm{R}_x(w)$ that $v = w$.

Second, to show that $\mathcal{V}$ is open and $E\colon \mathcal{T} \to \mathcal{V}$ is a diffeomorphism, it remains to check that the differential of E is invertible everywhere in $\mathcal{T}$ (the result then follows from applying the inverse function theorem at each point of $\mathcal{T}$, see Theorem 4.16). To this end, consider any $(x, v) \in \mathcal{T}$. Somewhat informally, the differential $\mathrm{D}E(x, v)$ is a block matrix of size two-by-two as displayed below. (To be formal, we should give a more precise description of the tangent space to $\mathrm{T}\mathcal{M}$ at (x, v). Here, it is identified with $\mathrm{T}_x\mathcal{M} \times \mathrm{T}_x\mathcal{M}$.)

$$\mathrm{D}E(x, v) \simeq \begin{bmatrix} I & 0 \\ * & \mathrm{DR}_x(v) \end{bmatrix}$$

Indeed, the differential of the first entry of $E(x, v) = (x, \mathrm{R}_x(v))$ with respect to x is the identity, and it is zero with respect to v. The second entry has some unspecified differential with respect to x, while its differential with respect to v is $\mathrm{DR}_x(v)$. Crucially, since v is in $\mathcal{T}_x$, we know by assumption that R_x is a diffeomorphism around v, hence $\mathrm{DR}_x(v)$ is invertible. We conclude that $\mathrm{D}E(x, v)$ is invertible for all $(x, v) \in \mathcal{T}$, as announced. □

In particular, under the stated conditions, $(x, y) \mapsto (x, \mathrm{R}_x^{-1}(y))$ is a diffeomorphism from $\mathcal{V}$ to $\mathcal{T}$, meaning the inverse retraction can be defined smoothly jointly in x and y (with care when it comes to domains).

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In this last proposition, the fact that $\mathcal{T}$ is open is crucial: this is what ties the domains $\mathcal{T}_x$ together. Without this assumption, we can still have an inverse, but not necessarily a smooth inverse. Furthermore, it is natural to want to include the tangent space origins in $\mathcal{T}$, that is, to make $\mathcal{T}$ a neighborhood of the zero section in $\mathrm{T}\mathcal{M}$. It is convenient to make this happen using a continuous function $\Delta\colon \mathcal{M} \to (0, \infty]$:

$$\mathcal{T} = \big\{(x, v) \in \mathrm{T}\mathcal{M} : \|v\|_x < \Delta(x)\big\}. \tag{10.8}$$

If R_x is defined and is a diffeomorphism on the open ball $B(x, \Delta(x))$ in $\mathrm{T}_x\mathcal{M}$ for all x, then Proposition 10.23 applies, and $\mathcal{V}$ contains the diagonal $\{(x, x) : x \in \mathcal{M}\}$.

Conveniently, for the exponential map, we can take Δ to be as large as one could possibly hope, namely: we can choose Δ to be the injectivity radius function. (This holds even if $\mathcal{M}$ is not connected or complete, see Section 10.8.)

Proposition 10.24. *On a Riemannian manifold $\mathcal{M}$, the injectivity radius function* $\mathrm{inj}\colon \mathcal{M} \to (0, \infty]$ *is continuous.*

Corollary 10.25. *The map $(x, v) \mapsto (x, \mathrm{Exp}_x(v))$ is a diffeomorphism from*

$$\mathcal{T} = \big\{(x, v) \in \mathrm{T}\mathcal{M} : \|v\|_x < \mathrm{inj}(x)\big\}$$

to

$$\mathcal{V} = \big\{(x, y) \in \mathcal{M} \times \mathcal{M} : \mathrm{dist}(x, y) < \mathrm{inj}(x)\big\}.$$

Its inverse is $(x, y) \mapsto (x, \mathrm{Log}_x(y))$, smooth from $\mathcal{V}$ to $\mathcal{T}$.

Under this corollary, we see that $(x, y) \mapsto \mathrm{Log}_x(y)$ is smooth jointly in x and y over some domain. This is apparent in Example 10.21 for the sphere, where $\mathrm{inj}(x) = \pi$ for all x.

More generally, we show that for any retraction there exists a positive and continuous function Δ which can be used to argue existence of a smooth inverse.

Proposition 10.26. *On a Riemannian manifold $\mathcal{M}$, consider the following open subsets of the tangent bundle $\mathrm{T}\mathcal{M}$:*

$$V_\delta(x) = \{(x', v') \in \mathrm{T}\mathcal{M} : \mathrm{dist}(x, x') < \delta \text{ and } \|v'\|_{x'} < \delta\}.$$

(In particular, x and x' must be in the same connected component.) Notice that $(x, 0)$ is in $V_\delta(x)$ for all $\delta > 0$. For any retraction R on $\mathcal{M}$ defined on a neighborhood $\mathcal{O}$ of the zero section in $\mathrm{T}\mathcal{M}$, define $\Delta\colon \mathcal{M} \to (0, \infty]$ by:

$$\Delta(x) = \sup\big\{\delta > 0 : V_\delta(x) \subseteq \mathcal{O} \text{ and } E \text{ is a diffeomorphism on } V_\delta(x)\big\},$$

where E is as defined in (10.7). Then, Δ is positive and continuous, and R_x is defined and is a diffeomorphism on $B(x, \Delta(x))$ for all x.

Proof. To see that $\Delta(x)$ is positive at every x, apply the inverse function theorem

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to the fact that the differential of E at $(x, 0)$ is invertible, since it is of the form $\mathrm{D}E(x, 0) \simeq \left[\begin{smallmatrix} I & 0 \\ * & I \end{smallmatrix}\right]$ (same as in the proof of Proposition 10.23).

It is sufficient to reason on each connected component of $\mathcal{M}$ separately, hence we may assume $\mathcal{M}$ is connected. If $\Delta(x) = \infty$ for some x, then $\mathcal{O} = \mathrm{T}\mathcal{M}$ and E is a diffeomorphism on that domain, so that $\Delta(x) = \infty$ for all x: this is compatible with the claim. We can now assume $\Delta(x)$ is finite for all x.

To see that Δ is continuous, we show that $\Delta(x) - \Delta(x') \leq \operatorname{dist}(x, x')$ for every two points $x, x' \in \mathcal{M}$. Then, switching the roles of x and x', we find $|\Delta(x) - \Delta(x')| \leq \operatorname{dist}(x, x')$, which shows Δ is continuous with respect to the Riemannian distance. This is equivalent to continuity with respect to the atlas topology by Theorem 10.3.

Pick any two points $x, x' \in \mathcal{M}$. If $\operatorname{dist}(x, x') \geq \Delta(x)$, the claim is clear. So assume $\operatorname{dist}(x, x') < \Delta(x)$, and define $\delta = \Delta(x) - \operatorname{dist}(x, x')$. We claim that

$$V_\delta(x') \subset V_{\Delta(x)}(x).$$

Indeed, pick any $(x'', v'') \in V_\delta(x')$. Then,

1. $\|v''\|_{x''} < \delta \leq \Delta(x)$, and
2. $\operatorname{dist}(x'', x) \leq \operatorname{dist}(x'', x') + \operatorname{dist}(x', x) < \delta + \operatorname{dist}(x', x) = \Delta(x)$.

We know E is a diffeomorphism on $V_{\Delta(x)}(x)$. Thus, E is also a diffeomorphism on $V_\delta(x')$. By definition of $\Delta(x')$, this implies $\Delta(x') \geq \delta = \Delta(x) - \operatorname{dist}(x, x')$, which is what we needed to show.

The conclusion about R_x follows from the fact that E is a diffeomorphism on each $V_{\Delta(x)}(x)$ (which covers $B(x, \Delta(x))$) and from the form of $\mathrm{D}E$, as in the proof of Proposition 10.23. □

For general retractions, we obtain the following corollary which notably means that, over some domain of $\mathcal{M} \times \mathcal{M}$ which contains all pairs (x, x), the map $(x, y) \mapsto \mathrm{R}_x^{-1}(y)$ can be defined smoothly jointly in x and y. There is no need to require that $\mathcal{M}$ be a Riemannian manifold because the existence of a Riemannian metric is guaranteed [Lee12, Prop. 13.3]: that is sufficient to apply Proposition 10.26.

Corollary 10.27. *For any retraction* R *on a manifold* $\mathcal{M}$ *there exists a neighborhood* $\mathcal{T}$ *of the zero section of the tangent bundle* $\mathrm{T}\mathcal{M}$ *on which*

$$(x, v) \mapsto (x, \mathrm{R}_x(v))$$

is a diffeomorphism; $\mathcal{T}$ *can be taken of the form* (10.8) *(with respect to an arbitrary Riemannian metric) with* $\Delta \colon \mathcal{M} \to (0, \infty]$ *continuous.*

We close with the notion of injectivity radius of a whole manifold. It may be zero, positive or infinite. The set of manifolds with positive injectivity radius is strictly included in the set of complete manifolds: Exercise 10.30. See [Lee18, Lem. 6.16] for a proof of Proposition 10.29.

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Definition 10.28. *The injectivity radius* $\mathrm{inj}(\mathcal{M})$ *of a Riemannian manifold* $\mathcal{M}$ *is the infimum of* $\mathrm{inj}(x)$ *over* $x \in \mathcal{M}$.

Proposition 10.29. *For a compact Riemannian manifold,* $\mathrm{inj}(\mathcal{M}) \in (0, \infty)$.

A Euclidean space has infinite injectivity radius. The unit sphere S^{n-1} has injectivity radius π. Importantly, the manifold $\mathbb{R}_r^{m\times n}$ of matrices with fixed rank r embedded in $\mathbb{R}^{m\times n}$ (Section 7.5) has *zero* injectivity radius. This is in part due to the fact that there exist matrices in $\mathbb{R}_r^{m\times n}$ that are arbitrarily close to matrices of rank strictly less than r, as measured in the embedding space ($\mathbb{R}_r^{m\times n}$ is not complete).

Exercise 10.30. *Show that if* $\mathrm{inj}(\mathcal{M})$ *is positive then* $\mathcal{M}$ *is complete. The converse is not true: give an example of a complete, connected manifold whose injectivity radius is zero.*

Exercise 10.31. *Let* $\mathcal{K}$ *be any subset of a Riemannian manifold* $\mathcal{M}$ *and let* $r\colon \mathcal{K} \to \mathbb{R}^+$ *be continuous, with* $\mathbb{R}^+ = \{t \in \mathbb{R} : t \geq 0\}$. *Show that*

$$\mathcal{T} = \{(x, s) \in \mathrm{T}\mathcal{M} : x \in \mathcal{K} \text{ and } \|s\|_x \leq r(x)\}$$

is compact in $\mathrm{T}\mathcal{M}$ *if and only if* $\mathcal{K}$ *is compact in* $\mathcal{M}$.

Exercise 10.32. *Let* $\mathcal{M} = \mathcal{M}_1 \times \mathcal{M}_2$ *be a Riemannian product manifold, as in Example 3.57. Let* $\mathcal{O}_1, \mathcal{O}_2$ *be the domains of the exponential maps on* $\mathcal{M}_1, \mathcal{M}_2$, *respectively. Show that the domain of the exponential map on* $\mathcal{M}$ *is* $\mathcal{O} = \mathcal{O}_1 \times \mathcal{O}_2$, *with the identification of tangent bundles on product manifolds as in* (3.31). *For* $(x_1, v_1) \in \mathcal{O}_1$ *and* $(x_2, v_2) \in \mathcal{O}_2$, *show that*

$$\mathrm{Exp}_x(v) = \big(\mathrm{Exp}_{x_1}(v_1), \mathrm{Exp}_{x_2}(v_2)\big)\,,$$

where $x = (x_1, x_2)$ *and* $v = (v_1, v_2)$, *and* Exp *denotes the exponential map on each respective manifold. Hint: use Exercise 5.39.*

10.3 Parallel transport

Consider a manifold $\mathcal{M}$ equipped with a connection ∇, and a tangent vector $u \in \mathrm{T}_x\mathcal{M}$.[3] In several situations, it is desirable to somehow transport u from x to another point $y \in \mathcal{M}$. In so doing, we would like for u and its transported version to be related in some meaningful way.

The geometric tool of choice for this task is called *parallel transport* (or *parallel translation*). Let $c\colon I \to \mathcal{M}$ be a smooth curve such that

$$c(0) = x \qquad \text{and} \qquad c(1) = y.$$

Consider a smooth vector field $Z \in \mathfrak{X}(c)$ on this curve with $Z(0) = u$. If Z does

[3] In this section, $\mathcal{M}$ may or may not be Riemannian, and ∇ may or may not be the Riemannian connection.

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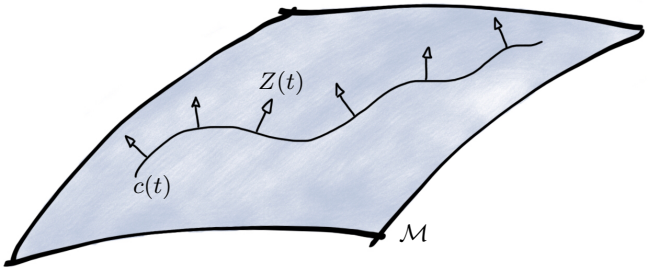


Figure 10.1 Parallel transports 'move' tangent vectors from one tangent space to another, along a specified curve. They can be used to compare or combine tangent vectors at different points by transporting them to a common tangent space.

not 'vary' too much, it is tempting to consider $Z(1)$ as a transport of u to y. One convenient way to formalize this is to require that Z be *parallel* with respect to the chosen connection ∇. Explicitly, using the covariant derivative $\frac{\mathrm{D}}{\mathrm{d}t}$ induced by ∇ (Theorem 5.29), we require

$$Z \in \mathfrak{X}(c), \qquad Z(0) = u, \qquad \text{and} \qquad \frac{\mathrm{D}}{\mathrm{d}t} Z = 0. \tag{10.9}$$

Using standard tools from linear ordinary differential equations, one can show that such a vector field exists and is unique [Lee18, Thm. 4.32].

Definition 10.33. *A vector field $Z \in \mathfrak{X}(c)$ such that $\frac{\mathrm{D}}{\mathrm{d}t} Z = 0$ is* parallel.

Theorem 10.34. *On a manifold $\mathcal{M}$ with a connection and induced covariant derivative $\frac{\mathrm{D}}{\mathrm{d}t}$, for any smooth curve $c\colon I \to \mathcal{M}$, $t_0 \in I$ and $u \in \mathrm{T}_{c(t_0)}\mathcal{M}$, there exists a unique parallel vector field $Z \in \mathfrak{X}(c)$ such that $Z(t_0) = u$.*

This justifies the following definition of parallel transport (Figure 10.1).

Definition 10.35. *Given a smooth curve c on $\mathcal{M}$, the* parallel transport *of tangent vectors at $c(t_0)$ to the tangent space at $c(t_1)$ along c is the map*

$$\mathrm{PT}^c_{t_1 \leftarrow t_0} \colon \mathrm{T}_{c(t_0)}\mathcal{M} \to \mathrm{T}_{c(t_1)}\mathcal{M}$$

defined by $\mathrm{PT}^c_{t_1 \leftarrow t_0}(u) = Z(t_1)$, where $Z \in \mathfrak{X}(c)$ is the unique parallel vector field such that $Z(t_0) = u$.

In particular, $t \mapsto \mathrm{PT}^c_{t \leftarrow t_0}(u)$ is a parallel vector field along c. Also, if Z is parallel along c, then $Z(t_1) = \mathrm{PT}^c_{t_1 \leftarrow t_0}(Z(t_0))$ for all t_0, t_1 in the domain of c.

On occasion, we may write $\mathrm{PT}^c_{y \leftarrow x}$ or even $\mathrm{PT}_{y \leftarrow x}$ when the times t_0, t_1 and the curve c such that $x = c(t_0)$ and $y = c(t_1)$ are clear from context, but beware: even if we use the Riemannian connection,

Parallel transport from x to y depends on the choice of curve connecting x and y.

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Indeed, think of a tangent vector at the equator pointing North. Transport it to the North pole via the shortest path. Alternatively, transport the same vector by first moving along the equator for some distance before going to the North pole: the results are different. This is, in fact, a crucial feature of Riemannian geometry, intimately related to the notion of curvature [Lee18, Ch. 7].

On a Riemannian manifold, when the curve c is not specified, one often implicitly means to move along the minimizing geodesic connecting x and y, assuming it exists and is unique.

Proposition 10.36. *The parallel transport operator* $\mathrm{PT}^c_{t_1\leftarrow t_0}$ *is linear. Also,* $\mathrm{PT}^c_{t_2\leftarrow t_1} \circ \mathrm{PT}^c_{t_1\leftarrow t_0} = \mathrm{PT}^c_{t_2\leftarrow t_0}$ *and* $\mathrm{PT}^c_{t\leftarrow t}$ *is the identity. In particular, the inverse of* $\mathrm{PT}^c_{t_1\leftarrow t_0}$ *is* $\mathrm{PT}^c_{t_0\leftarrow t_1}$*. If* $\mathcal{M}$ *is Riemannian and* ∇ *is compatible with the Riemannian metric, then*[4] *parallel transport is an isometry, that is,*

$$\forall u, v \in \mathrm{T}_{c(t_0)}\mathcal{M}, \qquad \langle u, v\rangle_{c(t_0)} = \left\langle \mathrm{PT}^c_{t_1\leftarrow t_0}(u), \mathrm{PT}^c_{t_1\leftarrow t_0}(v)\right\rangle_{c(t_1)}.$$

Stated differently, the adjoint and the inverse of $\mathrm{PT}^c_{t_1\leftarrow t_0}$ *coincide.*

Proof. For linearity, consider $u, v \in \mathrm{T}_{c(t_0)}\mathcal{M}$ and $a, b \in \mathbb{R}$, arbitrary. By Theorem 10.34, there exist unique parallel vector fields $Z_u, Z_v \in \mathfrak{X}(c)$ such that $Z_u(t_0) = u$ and $Z_v(t_0) = v$. Since $Z = aZ_u + bZ_v \in \mathfrak{X}(c)$ is also parallel and $Z(t_0) = au + bv$, we conclude that Z is the unique parallel vector field used in the definition of

$$\begin{aligned}\mathrm{PT}^c_{t_1\leftarrow t_0}(au + bv) = Z(t_1) &= aZ_u(t_1) + bZ_v(t_1) \\ &= a\mathrm{PT}^c_{t_1\leftarrow t_0}(u) + b\mathrm{PT}^c_{t_1\leftarrow t_0}(v),\end{aligned}$$

which shows linearity. The composition rule is clear, as is the fact that $\mathrm{PT}^c_{t\leftarrow t}$ is the identity. Then, the inverse follows by setting $t_2 = t_0$ in the composition rule.

To verify isometry, notice that

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle Z_u(t), Z_v(t)\rangle_{c(t)} = \left\langle \frac{\mathrm{D}}{\mathrm{d}t}Z_u(t), Z_v(t)\right\rangle_{c(t)} + \left\langle Z_u(t), \frac{\mathrm{D}}{\mathrm{d}t}Z_v(t)\right\rangle_{c(t)} = 0,$$

using compatibility of the covariant derivative with the Riemannian metric and the fact that Z_u, Z_v are parallel. Thus, the inner product is constant along c. □

One convenient tool afforded to us by parallel transports is the notion of *parallel frames* along a curve c. Consider an arbitrary basis $e_1, \ldots, e_d$ for the tangent space at $c(\bar{t})$, for an arbitrary $\bar{t}$ in the domain of definition of c. Construct the parallel vector fields

$$E_i(t) = \mathrm{PT}^c_{t\leftarrow\bar{t}}(e_i), \qquad i = 1, \ldots, d. \tag{10.10}$$

Since parallel transports are invertible, for all t, the vectors $E_i(t)$ form a basis for the tangent space at $c(t)$. (Also, if the manifold is Riemannian and we transport via a connection that is compatible with the metric, then orthonormality would

[4] The converse also holds: if parallel transport is an isometry, then ∇ is compatible with the metric [Lee18, Prop. 5.5].

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be preserved.) As a result, for any $Y \in \mathfrak{X}(c)$ there exist unique, real functions α_i such that

$$Y(t) = \sum_{i=1}^{d} \alpha_i(t) E_i(t). \tag{10.11}$$

These functions are smooth since Y is smooth. Owing to linearity,

$$\mathrm{PT}^c_{t_1 \leftarrow t_0}(Y(t_0)) = \sum_{i=1}^{d} \alpha_i(t_0) E_i(t_1).$$

In particular, we see that this is smooth in both t_0 and t_1.

Using parallel frames, we can show that covariant derivatives admit a convenient expression in terms of parallel transports: transport the vector field to a common tangent space, differentiate in the usual way (in that fixed tangent space), then transport back.

Proposition 10.37. *Consider a smooth curve $c\colon I \to \mathcal{M}$. Given a vector field $Z \in \mathfrak{X}(c)$ and $t_0 \in I$, let $z\colon I \to \mathrm{T}_{c(t_0)}\mathcal{M}$ be $z(t) = \mathrm{PT}^c_{t_0 \leftarrow t} Z(t)$. Then,*

$$\frac{\mathrm{D}}{\mathrm{d}t} Z(t) = \mathrm{PT}^c_{t \leftarrow t_0}\left(\frac{\mathrm{d}}{\mathrm{d}t} z(t)\right) = \lim_{\delta \to 0} \frac{\mathrm{PT}^c_{t \leftarrow t+\delta} Z(t+\delta) - Z(t)}{\delta}.$$

Proof. Transport any basis $e_1, \ldots, e_d$ of $\mathrm{T}_{c(t_0)}\mathcal{M}$ along c to form a frame $E_i(t) = \mathrm{PT}^c_{t \leftarrow t_0} e_i$. There exist unique, smooth, real functions $\alpha_1, \ldots, \alpha_d$ such that $Z(t) = \sum_{i=1}^{d} \alpha_i(t) E_i(t)$. Then, by the properties of covariant derivatives (Theorem 5.29) and $\frac{\mathrm{D}}{\mathrm{d}t} E_i = 0$,

$$\begin{aligned}
\frac{\mathrm{D}}{\mathrm{d}t} Z(t) &= \sum_{i=1}^{d} \alpha_i'(t) E_i(t) \\
&= \mathrm{PT}^c_{t \leftarrow t_0} \sum_{i=1}^{d} \alpha_i'(t) e_i \\
&= \mathrm{PT}^c_{t \leftarrow t_0} \frac{\mathrm{d}}{\mathrm{d}t} \sum_{i=1}^{d} \alpha_i(t) e_i = \mathrm{PT}^c_{t \leftarrow t_0} \frac{\mathrm{d}}{\mathrm{d}t} z(t),
\end{aligned}$$

as announced. The important point is that z is a map between (fixed) linear spaces, which is why we can take a classical derivative $\frac{\mathrm{d}}{\mathrm{d}t}$. □

Exercise 10.38. *On the sphere S^{n-1} as a Riemannian submanifold of $\mathbb{R}^n$ with the usual metric and connection, parallel transport along the geodesic $c(t) = \mathrm{Exp}_x(tv)$ admits the following explicit expression [QGA10a]:*

$$\mathrm{PT}^c_{t \leftarrow 0}(u) = \left(I_n + (\cos(t\|v\|) - 1)\frac{vv^\top}{\|v\|^2} - \sin(t\|v\|)\frac{xv^\top}{\|v\|}\right) u.$$

Verify this claim. (Recall Example 5.37 for the exponential.)

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Exercise 10.39. *Let $c = (c_1, c_2)$ be a smooth curve on a Riemannian product manifold $\mathcal{M} = \mathcal{M}_1 \times \mathcal{M}_2$. Verify that parallel transport along c is given in terms of parallel transports along c_1 and c_2 as*

$$\mathrm{PT}^{c}_{t_b \leftarrow t_a}(v) = \left(\mathrm{PT}^{c_1}_{t_b \leftarrow t_a}(v_1), \mathrm{PT}^{c_2}_{t_b \leftarrow t_a}(v_2)\right),$$

where $v = (v_1, v_2)$ is tangent to $\mathcal{M}$ at $c(t_a)$. Hint: recall Exercise 5.34.

10.4 Lipschitz conditions and Taylor expansions

One of the most convenient regularity assumptions one can make regarding the cost function f of an optimization problem is that it or its derivatives be Lipschitz continuous. Indeed, in the Euclidean case, it is well known that such properties lead to global bounds on the discrepancy between f and its Taylor expansions of various orders. These, in turn, ease worst-case iteration complexity analyses. Here, we consider definitions of Lipschitz continuity on Riemannian manifolds, and we derive Taylor bounds analogous to their Euclidean counterparts. In so doing, we are careful not to require the manifold to be complete.

⋆ In this section,

1. We implicitly assume all of our manifolds are Riemannian,
2. We usually omit subscripts for inner products and norms (writing $\langle \cdot, \cdot \rangle$ and $\| \cdot \|$ instead of $\langle \cdot, \cdot \rangle_x$ and $\| \cdot \|_x$), and
3. We state explicitly how many times we need maps to be (continuously) differentiable, thus not assuming smoothness (infinite differentiability) by default: recall Remark 8.6.

Let A, B be two metric spaces. A map $F \colon A \to B$ is *L-Lipschitz continuous* if $L \geq 0$ is such that

$$\forall x, y \in A, \qquad \mathrm{dist}_B(F(x), F(y)) \leq L \, \mathrm{dist}_A(x, y), \tag{10.12}$$

where dist_A, dist_B denote the distances on A and B. In particular:

Definition 10.40. *A function $f \colon \mathcal{M} \to \mathbb{R}$ on a connected manifold $\mathcal{M}$ is* L-Lipschitz continuous *if*

$$\forall x, y \in \mathcal{M}, \qquad |f(x) - f(y)| \leq L \, \mathrm{dist}(x, y), \tag{10.13}$$

where dist *is the Riemannian distance on $\mathcal{M}$. If $\mathcal{M}$ is disconnected, we require the condition to hold on each connected component separately.*

The definition above can be reformulated as we show below. This second formulation is more convenient to study iterates of optimization algorithms presented as $x_{k+1} = \mathrm{Exp}_{x_k}(s_k)$ for some $s_k \in \mathrm{T}_{x_k}\mathcal{M}$.

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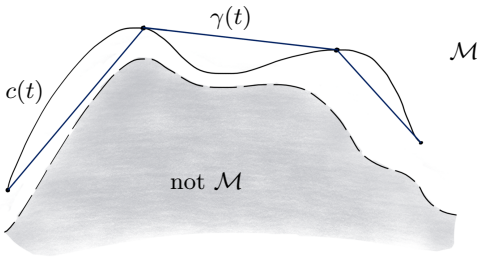


Figure 10.2 In Lemma 10.42, the curve γ is made of a finite number of minimizing geodesic segments, with endpoints on c.

Proposition 10.41. *A function $f : \mathcal{M} \to \mathbb{R}$ is L-Lipschitz continuous if and only if*

$$\forall (x, s) \in \mathcal{O}, \qquad |f(\mathrm{Exp}_x(s)) - f(x)| \leq L\|s\|, \tag{10.14}$$

where $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$ is the domain of the exponential map (Definition 10.16).

To prove this, we first introduce a lemma which states that any continuous curve c can be interpolated by a 'broken geodesic' γ. Also, if c is piecewise smooth it has a length and we have $L(\gamma) \leq L(c)$. See Figure 10.2.

Lemma 10.42. *Given $c : [0, 1] \to \mathcal{M}$ continuous on a manifold $\mathcal{M}$, there exist a finite number of times $0 = t_0 < t_1 < \cdots < t_{n-1} < t_n = 1$ such that $\mathrm{dist}(c(t_i), c(t_{i+1})) < \mathrm{inj}(c(t_i))$ for $i = 0, \ldots, n-1$.*

These times define a piecewise regular curve $\gamma : [0, 1] \to \mathcal{M}$ satisfying $\gamma(t_i) = c(t_i)$ and such that $\gamma|_{[t_i, t_{i+1}]}$ is the minimizing geodesic connecting its endpoints. As such, there exist tangent vectors $s_0, \ldots, s_{n-1}$ such that $\gamma(t_{i+1}) = \mathrm{Exp}_{\gamma(t_i)}(s_i)$ and $\sum_{i=0}^{n-1} \|s_i\| = L(\gamma)$.

Proof. Consider the recursive routine `construct` with inputs a, b which proceeds as follows: if $\mathrm{dist}(c(a), c(b)) < \mathrm{inj}(c(a))$, return (a, b); if not, return the results of `construct`$(a, (a + b)/2)$ and `construct`$((a + b)/2, b)$ merged. We claim that `construct`$(0, 1)$ is an appropriate selection. Indeed, the routine terminates after a finite number of steps because $\mathrm{inj} \circ c$ is continuous and positive on the compact domain $[0, 1]$ so that it is bounded away from zero, and c is continuous so that for $\varepsilon > 0$ small enough we can have $\mathrm{dist}(c(t), c(t+\varepsilon))$ arbitrarily small. Furthermore, for all selected t_i, t_{i+1}, we have $\mathrm{dist}(c(t_i), c(t_{i+1})) < \mathrm{inj}(c(t_i))$. Hence, there exists a (unique) minimizing geodesic connecting $c(t_i)$ to $c(t_{i+1})$, for all i, and $\|s_i\| = L(\gamma|_{[t_i, t_{i+1}]})$. (We used Proposition 10.22.) □

Proof of Proposition 10.41. Under condition (10.13), the claim is clear:

$$\forall (x, s) \in \mathcal{O}, \qquad |f(\mathrm{Exp}_x(s)) - f(x)| \leq L\,\mathrm{dist}(\mathrm{Exp}_x(s), x) \leq L\|s\|$$

since $t \mapsto \mathrm{Exp}_x(ts)$ is a smooth curve defined on $[0, 1]$ with length $\|s\|$.

The other way around, let us assume condition (10.14) holds. If $\mathcal{M}$ is complete

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(that is, $\mathcal{O} = \mathrm{T}\mathcal{M}$) the claim is also clear: by Theorem 10.9, for all $x, y \in \mathcal{M}$ in the same connected component there exists s in $\mathrm{T}_x\mathcal{M}$ such that $y = \mathrm{Exp}_x(s)$ and $\|s\| = \mathrm{dist}(x, y)$.

If $\mathcal{M}$ is not complete, we proceed as follows: by definition of distance (10.2), for all $x, y \in \mathcal{M}$ in the same connected component and for all $\varepsilon > 0$, there exists a piecewise regular curve $c\colon [0, 1] \to \mathcal{M}$ with length $L(c) \leq \mathrm{dist}(x, y) + \varepsilon$ such that $c(0) = x$ and $c(1) = y$. Construct a broken geodesic γ as provided by Lemma 10.42: there exist times $0 = t_0 < \cdots < t_n = 1$ and tangent vectors $s_0, \ldots, s_{n-1}$ such that $\gamma(t_i) = c(t_i)$ and $\gamma(t_{i+1}) = \mathrm{Exp}_{\gamma(t_i)}(s_i)$ for all i, and $\sum_{i=0}^{n-1} \|s_i\| = L(\gamma) \leq L(c)$. Then,

$$|f(x) - f(y)| \leq \sum_{i=0}^{n-1} |f(\gamma(t_i)) - f(\gamma(t_{i+1}))| \overset{(10.14)}{\leq} \sum_{i=0}^{n-1} L \cdot \|s_i\| \leq L \cdot L(c).$$

This reasoning holds for all $\varepsilon > 0$, hence condition (10.13) follows. □

If f has a continuous gradient, then f is Lipschitz continuous exactly if its gradient is bounded.

Proposition 10.43. *If $f\colon \mathcal{M} \to \mathbb{R}$ has a continuous gradient, then f is L-Lipschitz continuous if and only if*

$$\forall x \in \mathcal{M}, \qquad \|\mathrm{grad} f(x)\| \leq L. \tag{10.15}$$

Proof. For any $(x, s) \in \mathcal{O}$, consider $c(t) = \mathrm{Exp}_x(ts)$ for $t \in [0, 1]$. Then,

$$f(c(1)) - f(c(0)) = \int_0^1 (f \circ c)'(t) \mathrm{d}t = \int_0^1 \langle \mathrm{grad} f(c(t)), c'(t) \rangle \, \mathrm{d}t.$$

Thus, if the gradient norm is bounded by L at all points along c,

$$|f(\mathrm{Exp}_x(s)) - f(x)| \leq L \int_0^1 \|c'(t)\| \mathrm{d}t = L \cdot L(c) = L\|s\|.$$

This shows that (10.14) holds.

The other way around, for any $x \in \mathcal{M}$, assuming (10.14) holds and using that the domain of Exp_x is open around the origin, we have:

$$\begin{aligned} \|\mathrm{grad} f(x)\| &= \max_{s \in \mathrm{T}_x\mathcal{M}, \|s\|=1} \langle \mathrm{grad} f(x), s \rangle \\ &= \max_{s \in \mathrm{T}_x\mathcal{M}, \|s\|=1} \mathrm{D}f(x)[s] \\ &= \max_{s \in \mathrm{T}_x\mathcal{M}, \|s\|=1} \lim_{t \to 0^+} \frac{f(\mathrm{Exp}_x(ts)) - f(x)}{t} \leq L, \end{aligned}$$

since $f(\mathrm{Exp}_x(ts)) - f(x) \leq L\|ts\| = L|t|$. □

We now turn to defining Lipschitz continuity for the Riemannian gradient of a function f. Since $\mathrm{grad} f$ is a map from $\mathcal{M}$ to $\mathrm{T}\mathcal{M}$, to apply the general notion of Lipschitz continuity directly we would need to pick a distance on the tangent bundle. However, this would not lead to interesting notions for us. Indeed, the

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distance between $\operatorname{grad} f(x)$ and $\operatorname{grad} f(y)$ would necessarily have to be positive if $x \neq y$ since they would always be distinct points in $\mathrm{T}\mathcal{M}$. Contrast this to the Euclidean case $f : \mathcal{E} \to \mathbb{R}$, where it is natural to measure $\operatorname{grad} f(x) - \operatorname{grad} f(y)$ in the Euclidean metric, disregarding the base points. With this in mind, it is reasonable to resort to parallel transport (Section 10.3) to compare tangent vectors at distinct points. Since parallel transport is dependent on paths, this leaves some leeway in the definition.

The following definition is fairly common. Notice how the restriction by the injectivity radius allows us to choose a privileged path along which to transport (owing to Proposition 10.22). For our purpose, the notion below is particularly relevant with $V = \operatorname{grad} f$, in which case we would say f has an *L-Lipschitz continuous gradient*.

Definition 10.44. *A vector field V on a connected manifold $\mathcal{M}$ is* L-Lipschitz continuous *if, for all $x, y \in \mathcal{M}$ with $\operatorname{dist}(x, y) < \operatorname{inj}(x)$,*

$$\|\mathrm{PT}^{\gamma}_{0\leftarrow 1} V(y) - V(x)\| \leq L \operatorname{dist}(x, y), \tag{10.16}$$

where $\gamma : [0, 1] \to \mathcal{M}$ is the unique minimizing geodesic connecting x to y. If $\mathcal{M}$ is disconnected, we require the condition on each connected component.

Here too, we provide an equivalent definition in terms of the exponential map: this may be more convenient to analyze optimization algorithms, and has the added benefit of allowing the comparison of points which are further apart than the injectivity radius (but still connected by a geodesic).

Proposition 10.45. *A vector field V on a manifold $\mathcal{M}$ is L-Lipschitz continuous if and only if*

$$\forall (x, s) \in \mathcal{O}, \qquad \|P_s^{-1} V(\operatorname{Exp}_x(s)) - V(x)\| \leq L\|s\|, \tag{10.17}$$

where $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$ is the domain of Exp *and P_s denotes parallel transport along $\gamma(t) = \operatorname{Exp}_x(ts)$ from $t = 0$ to $t = 1$.*

Proof. For any $x, y \in \mathcal{M}$ such that $\operatorname{dist}(x, y) < \operatorname{inj}(x)$, there exists a unique $s \in \mathrm{T}_x\mathcal{M}$ such that $y = \operatorname{Exp}_x(s)$ and $\|s\| = \operatorname{dist}(x, y)$. Thus, if condition (10.17) holds, then (10.16) holds.

The other way around, for any $(x, s) \in \mathcal{O}$, consider the geodesic $\gamma(t) = \operatorname{Exp}_x(ts)$ defined over $[0, 1]$. It may or may not be minimizing. In any case, owing to Lemma 10.42, the interval $[0, 1]$ can be partitioned by $0 = t_0 < \cdots < t_n = 1$ such that $\operatorname{dist}(\gamma(t_i), \gamma(t_{i+1})) < \operatorname{inj}(\gamma(t_i))$. Since

$$\mathrm{PT}^{\gamma}_{0\leftarrow 1} = \mathrm{PT}^{\gamma}_{t_0 \leftarrow t_{n-1}} \circ \mathrm{PT}^{\gamma}_{t_{n-1} \leftarrow t_n}$$

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and since parallel transport is an isometry, we find that

$$\begin{aligned}\|\mathrm{PT}^{\gamma}_{t_0\leftarrow t_n}V(\gamma(t_n)) - V(x)\| &= \|\mathrm{PT}^{\gamma}_{t_{n-1}\leftarrow t_n}V(\gamma(t_n)) - \mathrm{PT}^{\gamma}_{t_{n-1}\leftarrow t_0}V(x)\| \\ &\leq \|\mathrm{PT}^{\gamma}_{t_{n-1}\leftarrow t_n}V(\gamma(t_n)) - V(\gamma(t_{n-1}))\| \\ &\quad + \|\mathrm{PT}^{\gamma}_{t_{n-1}\leftarrow t_0}V(x) - V(\gamma(t_{n-1}))\| \\ &\leq L\,\mathrm{dist}(\gamma(t_{n-1}),\gamma(t_n)) \\ &\quad + \|\mathrm{PT}^{\gamma}_{t_0\leftarrow t_{n-1}}V(\gamma(t_{n-1})) - V(x)\|,\end{aligned}$$

where in the last step we were able to use (10.16) since $\gamma|_{[t_{n-1},t_n]}$ is the unique minimizing geodesic connecting $\gamma(t_{n-1})$ and $\gamma(t_n)$. Repeat this argument on the right-most term $n-1$ times to see that

$$\|\mathrm{PT}^{\gamma}_{0\leftarrow 1}V(\mathrm{Exp}_x(s)) - V(x)\| \leq L\sum_{i=0}^{n-1}\mathrm{dist}(\gamma(t_i),\gamma(t_{i+1})) = L\cdot L(\gamma),$$

using that γ is a geodesic. To conclude, note that $L(\gamma) = \|s\|$ and that $\mathrm{PT}^{\gamma}_{0\leftarrow 1} = P_s^{-1}$, so that condition (10.17) holds. □

If the vector field V is continuously differentiable, then Lipschitz continuity of V is equivalent to boundedness of its covariant derivative. In turn, this makes it possible to compare the values of V at points connected by curves other than geodesics.

Proposition 10.46. *If V is a continuously differentiable vector field on a manifold $\mathcal{M}$, then it is L-Lipschitz continuous if and only if*

$$\forall (x,s)\in \mathrm{T}\mathcal{M}, \qquad \|\nabla_s V\| \leq L\|s\|, \tag{10.18}$$

where ∇ is the Riemannian connection. In that case, for any smooth curve $c\colon [0,1]\to\mathcal{M}$ connecting any x to any y, it holds that

$$\|\mathrm{PT}^{c}_{0\leftarrow 1}V(y) - V(x)\| \leq L\cdot L(c). \tag{10.19}$$

Proof. We first show that (10.18) implies (10.19). Since the latter itself implies (10.17), this also takes care of showing that (10.18) implies V is L-Lipschitz continuous. To this end, consider an orthonormal basis $e_1,\ldots,e_d\in \mathrm{T}_x\mathcal{M}$ and their parallel transports $E_i(t) = \mathrm{PT}^{c}_{t\leftarrow 0}(e_i)$. Then, $V(c(t)) = \sum_{i=1}^d v_i(t)E_i(t)$ for some continuously differentiable functions v_i, and

$$\sum_{i=1}^d v_i'(t)E_i(t) = \frac{\mathrm{D}}{\mathrm{d}t}(V\circ c)(t) = \nabla_{c'(t)}V.$$

Furthermore,

$$\begin{aligned}\mathrm{PT}^{c}_{0\leftarrow 1}V(c(1)) - V(c(0)) &= \sum_{i=1}^d (v_i(1)-v_i(0))e_i \\ &= \sum_{i=1}^d \left(\int_0^1 v_i'(t)\mathrm{d}t\right)e_i = \int_0^1 \mathrm{PT}^{c}_{0\leftarrow t}\big(\nabla_{c'(t)}V\big)\,\mathrm{d}t.\end{aligned}$$

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Consequently, using that parallel transports are isometric,

$$\|\mathrm{PT}^c_{0\leftarrow 1} V(y) - V(x)\| \leq \int_0^1 \|\nabla_{c'(t)} V\| \mathrm{d}t \overset{(10.18)}{\leq} L \int_0^1 \|c'(t)\| \mathrm{d}t = L \cdot L(c).$$

Now for the other direction: assume V is L-Lipschitz continuous. For any $x \in \mathcal{M}$, using that the domain of Exp_x is open around the origin, we know that for all $s \in \mathrm{T}_x\mathcal{M}$ the smooth curve $c(t) = \mathrm{Exp}_x(ts)$ is defined around $t = 0$. Then, by Proposition 10.37,

$$\nabla_s V = \frac{\mathrm{D}}{\mathrm{d}t} V(c(t))\bigg|_{t=0} = \lim_{t \to 0} \frac{\mathrm{PT}^c_{0\leftarrow t} V(c(t)) - V(c(0))}{t}.$$

By (10.17), the norm of the numerator is bounded by $L\|ts\|$, which concludes the proof. □

Corollary 10.47. *If $f \colon \mathcal{M} \to \mathbb{R}$ is twice continuously differentiable on a manifold $\mathcal{M}$, then* $\mathrm{grad} f$ *is L-Lipschitz continuous if and only if* $\mathrm{Hess} f(x)$ *has operator norm bounded by L for all x, that is, if for all x we have*

$$\|\mathrm{Hess} f(x)\| = \max_{\substack{s \in \mathrm{T}_x\mathcal{M} \\ \|s\| = 1}} \|\mathrm{Hess} f(x)[s]\| \leq L.$$

Let us summarize these findings.

Corollary 10.48. *For a vector field V on a manifold $\mathcal{M}$, these are equivalent:*

1. *V is L-Lipschitz continuous.*
2. *For all x, y in the same component with $\mathrm{dist}(x, y) < \mathrm{inj}(x)$, it holds that $\|\mathrm{PT}^\gamma_{0\leftarrow 1} V(y) - V(x)\| \leq L \,\mathrm{dist}(x, y)$ with γ the unique minimizing geodesic connecting x to y.*
3. *For all (x, s) in the domain of* Exp, *$\|P_s^{-1} V(\mathrm{Exp}_x(s)) - V(x)\| \leq L\|s\|$, where P_s is parallel transport along $c(t) = \mathrm{Exp}_x(ts)$ from $t = 0$ to $t = 1$.*

If V is continuously differentiable, the above are equivalent to the following:

4. *For all smooth $c \colon [0, 1] \to \mathcal{M}$, $\|\mathrm{PT}^c_{0\leftarrow 1} V(c(1)) - V(c(0))\| \leq L \cdot L(c)$.*
5. *For all $(x, s) \in \mathrm{T}\mathcal{M}$, $\|\nabla_s V\| \leq L\|s\|$.*

Particularizing the above to $V = \mathrm{grad} f$ provides a good understanding of functions f with Lipschitz continuous gradients.

Going one degree higher, we now define and (begin to) discuss functions with a *Lipschitz continuous Hessian*. The Hessian of f associates to each x a linear map $\mathrm{Hess} f(x)$ from $\mathrm{T}_x\mathcal{M}$ to itself. The following definition applies.

Definition 10.49. *For each $x \in \mathcal{M}$, let $H(x) \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ be linear. If $\mathcal{M}$ is connected, we say H is* L-Lipschitz continuous *if for all $x, y \in \mathcal{M}$ such that $\mathrm{dist}(x, y) < \mathrm{inj}(x)$ we have*

$$\|\mathrm{PT}^\gamma_{0\leftarrow 1} \circ H(y) \circ \mathrm{PT}^\gamma_{1\leftarrow 0} - H(x)\| \leq L \,\mathrm{dist}(x, y), \tag{10.20}$$

where $\|\cdot\|$ denotes the operator norm with respect to the Riemannian metric,

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and $\gamma\colon [0,1] \to \mathcal{M}$ is the unique minimizing geodesic connecting x to y. If $\mathcal{M}$ is disconnected, we require the condition on each connected component.

Recall the operator norm is $\|H(x)\| = \max_{s \in \mathrm{T}_x\mathcal{M}, \|s\|=1} \|H(x)[s]\|$. The proof of the following proposition is left as an exercise.

Proposition 10.50. *For each $x \in \mathcal{M}$, let $H(x)\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ be linear. The map H is L-Lipschitz continuous if and only if*

$$\forall (x,s) \in \mathcal{O}, \qquad \|P_s^{-1} \circ H(\mathrm{Exp}_x(s)) \circ P_s - H(x)\| \leq L\|s\|, \tag{10.21}$$

where $\mathcal{O} \subseteq \mathrm{T}\mathcal{M}$ is the domain of Exp and P_s denotes parallel transport along $\gamma(t) = \mathrm{Exp}_x(ts)$ from $t = 0$ to $t = 1$.

In Section 10.7, we define what it means for a map H as above to be differentiable, and we define its covariant derivative ∇H. Then, as a particular case of Proposition 10.83 we get the following claim, analogous to Proposition 10.46 above.

Proposition 10.51. *For each $x \in \mathcal{M}$, let $H(x)\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ be linear. If H is continuously differentiable, it is L-Lipschitz continuous if and only if*

$$\forall (x,s) \in \mathrm{T}\mathcal{M}, \qquad \|\nabla_s H\| \leq L\|s\|, \tag{10.22}$$

where ∇ is the Riemannian connection and $\nabla_s H\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M}$ is linear. In that case, for any smooth curve $c\colon [0,1] \to \mathcal{M}$ connecting x to y, we have

$$\|\mathrm{PT}^c_{0\leftarrow 1} \circ H(y) \circ \mathrm{PT}^c_{1\leftarrow 0} - H(x)\| \leq L \cdot L(c). \tag{10.23}$$

Corollary 10.52. *If $f\colon \mathcal{M} \to \mathbb{R}$ is three times continuously differentiable on a manifold $\mathcal{M}$, then $\mathrm{Hess} f$ is L-Lipschitz continuous if and only if*

$$\forall (x,s) \in \mathrm{T}\mathcal{M}, \qquad \|\nabla_s \mathrm{Hess} f\| \leq L\|s\|,$$

where $\nabla_s \mathrm{Hess} f$ is a self-adjoint linear map on $\mathrm{T}_x\mathcal{M}$ defined by (10.49).

A summary of the same kind as Corollary 10.48 holds here as well.

In Section 10.7, we show how to cast f, $\mathrm{grad} f$ and $\mathrm{Hess} f$ as tensor fields of order zero, one and two, respectively. We show how the Riemannian connection can be used to differentiate tensor fields in general, and we discuss Lipschitz continuity at that level of generality. This provides for the missing details in our brief discussion of Lipschitz continuous Hessians, and indicates how to deal with derivatives of arbitrary order.

We can now derive some of the most useful consequences of Lipschitz continuity, namely, bounds on the difference between a function f (or its derivatives) and corresponding Taylor expansions.

We use the following notation often: given (x,s) in the domain $\mathcal{O}$ of the exponential map, let $\gamma(t) = \mathrm{Exp}_x(ts)$ be the corresponding geodesic (defined in particular on the interval $[0,1]$); then, we let

$$P_{ts} = \mathrm{PT}^{\gamma}_{t\leftarrow 0} \tag{10.24}$$

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denote parallel transport from x to $\mathrm{Exp}_x(ts)$ along γ. Since γ is a geodesic, its velocity vector field is parallel and we have

$$\gamma'(t) = P_{ts}\gamma'(0) = P_{ts}s. \tag{10.25}$$

This will be helpful a number of times.

Proposition 10.53. *Let $f\colon \mathcal{M} \to \mathbb{R}$ be continuously differentiable on a manifold $\mathcal{M}$. Let $\gamma(t) = \mathrm{Exp}_x(ts)$ be defined on $[0,1]$ and assume there exists $L \geq 0$ such that, for all $t \in [0,1]$,*

$$\left\| P_{ts}^{-1}\mathrm{grad} f(\gamma(t)) - \mathrm{grad} f(x) \right\| \leq L\|ts\|.$$

Then, the following inequality holds:

$$|f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle| \leq \frac{L}{2}\|s\|^2.$$

Proof. Consider the real function $f \circ \gamma$ on $[0,1]$; we have:

$$\begin{aligned} f(\gamma(1)) &= f(\gamma(0)) + \int_0^1 (f\circ\gamma)'(t)\mathrm{d}t \\ &= f(x) + \int_0^1 \langle \mathrm{grad} f(\gamma(t)), \gamma'(t)\rangle \,\mathrm{d}t \\ &= f(x) + \int_0^1 \left\langle P_{ts}^{-1}\mathrm{grad} f(\gamma(t)), s\right\rangle \mathrm{d}t, \end{aligned}$$

where on the last line we used $\gamma'(t) = P_{ts}s$ (10.25) and the fact that P_{ts} is an isometry, so that its adjoint with respect to the Riemannian metric is equal to its inverse. Moving $f(x)$ to the left-hand side and subtracting $\langle \mathrm{grad} f(x), s\rangle$ on both sides, we get

$$f(\mathrm{Exp}_x(s)) - f(x) - \langle \mathrm{grad} f(x), s\rangle = \int_0^1 \left\langle P_{ts}^{-1}\mathrm{grad} f(\gamma(t)) - \mathrm{grad} f(x), s\right\rangle \mathrm{d}t.$$

Using Cauchy–Schwarz and our main assumption, it follows that

$$|f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle| \leq \int_0^1 tL\|s\|^2 \mathrm{d}t = \frac{L}{2}\|s\|^2,$$

as announced. □

The following corollary shows that the regularity assumptions A4.3 (p59) and A6.6 (p135) hold for the exponential retraction over its whole domain provided $\mathrm{grad} f$ is L-Lipschitz (Definition 10.44). See also Exercise 10.58.

Corollary 10.54. *If $f\colon \mathcal{M} \to \mathbb{R}$ has L-Lipschitz continuous gradient, then*

$$|f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle| \leq \frac{L}{2}\|s\|^2$$

for all (x, s) in the domain of the exponential map.

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Proposition 10.55. *Let $f\colon \mathcal{M} \to \mathbb{R}$ be twice continuously differentiable on a manifold $\mathcal{M}$. Let $\gamma(t) = \mathrm{Exp}_x(ts)$ be defined on $[0,1]$ and assume there exists $L \geq 0$ such that, for all $t \in [0,1]$,*

$$\left\| P_{ts}^{-1} \circ \mathrm{Hess} f(\gamma(t)) \circ P_{ts} - \mathrm{Hess} f(x) \right\| \leq L\|ts\|.$$

Then, the two following inequalities hold:

$$\left| f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x) \rangle - \frac{1}{2} \langle s, \mathrm{Hess} f(x)[s] \rangle \right| \leq \frac{L}{6}\|s\|^3,$$

$$\left\| P_s^{-1} \mathrm{grad} f(\mathrm{Exp}_x(s)) - \mathrm{grad} f(x) - \mathrm{Hess} f(x)[s] \right\| \leq \frac{L}{2}\|s\|^2.$$

Proof. The proof is in three steps.

Step 1: a preliminary computation. Pick an arbitrary basis $e_1, \ldots, e_d$ for $\mathrm{T}_x\mathcal{M}$ and define the parallel vector fields $E_i(t) = P_{ts}e_i$ along $\gamma(t)$. The vectors $E_1(t), \ldots, E_d(t)$ form a basis for $\mathrm{T}_{\gamma(t)}\mathcal{M}$ for each $t \in [0,1]$. As a result, we can express the gradient of f along $\gamma(t)$ in these bases,

$$\mathrm{grad} f(\gamma(t)) = \sum_{i=1}^{d} \alpha_i(t) E_i(t), \tag{10.26}$$

with $\alpha_1(t), \ldots, \alpha_d(t)$ differentiable. Using the Riemannian connection ∇ and associated covariant derivative $\frac{\mathrm{D}}{\mathrm{d}t}$, we find on the one hand that

$$\frac{\mathrm{D}}{\mathrm{d}t} \mathrm{grad} f(\gamma(t)) = \nabla_{\gamma'(t)} \mathrm{grad} f = \mathrm{Hess} f(\gamma(t))[\gamma'(t)],$$

and on the other hand that

$$\frac{\mathrm{D}}{\mathrm{d}t} \sum_{i=1}^{d} \alpha_i(t) E_i(t) = \sum_{i=1}^{d} \alpha_i'(t) E_i(t) = P_{ts} \sum_{i=1}^{d} \alpha_i'(t) e_i.$$

Combining with $\gamma'(t) = P_{ts}s$ (10.25), we deduce that

$$\sum_{i=1}^{d} \alpha_i'(t) e_i = \big(P_{ts}^{-1} \circ \mathrm{Hess} f(\gamma(t)) \circ P_{ts}\big)[s].$$

Going back to (10.26), we also see that

$$G(t) \triangleq P_{ts}^{-1} \mathrm{grad} f(\gamma(t)) = \sum_{i=1}^{d} \alpha_i(t) e_i$$

is a map from (a subset of) $\mathbb{R}$ to $\mathrm{T}_x\mathcal{M}$—two linear spaces—so that we can differentiate it in the usual way:

$$G'(t) = \sum_{i=1}^{d} \alpha_i'(t) e_i.$$

Overall, we conclude that

$$G'(t) = \frac{\mathrm{d}}{\mathrm{d}t} P_{ts}^{-1} \mathrm{grad} f(\gamma(t)) = \big(P_{ts}^{-1} \circ \mathrm{Hess} f(\gamma(t)) \circ P_{ts}\big)[s]. \tag{10.27}$$

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This comes in handy in the next step.

Step 2: Taylor expansion of the gradient. Since G' is continuous,

$$\begin{aligned} P_{ts}^{-1}\mathrm{grad}f(\gamma(t)) = G(t) &= G(0) + \int_0^t G'(\tau)\mathrm{d}\tau \\ &= \mathrm{grad}f(x) + \int_0^t \big(P_{\tau s}^{-1} \circ \mathrm{Hess}f(\gamma(\tau)) \circ P_{\tau s}\big)[s]\mathrm{d}\tau. \end{aligned}$$

Moving $\mathrm{grad}f(x)$ to the left-hand side and subtracting $\mathrm{Hess}f(x)[ts]$ on both sides, we find

$$\begin{aligned} &P_{ts}^{-1}\mathrm{grad}f(\gamma(t)) - \mathrm{grad}f(x) - \mathrm{Hess}f(x)[ts] \\ &\qquad = \int_0^t \big(P_{\tau s}^{-1} \circ \mathrm{Hess}f(\gamma(\tau)) \circ P_{\tau s} - \mathrm{Hess}f(x)\big)[s]\mathrm{d}\tau. \end{aligned}$$

Using the main assumption on $\mathrm{Hess}f$ along γ, it follows that

$$\begin{aligned} &\big\|P_{ts}^{-1}\mathrm{grad}f(\gamma(t)) - \mathrm{grad}f(x) - \mathrm{Hess}f(x)[ts]\big\| \\ &\qquad \leq \int_0^t \tau L\|s\|^2\mathrm{d}\tau = \frac{L}{2}\|ts\|^2. \end{aligned} \tag{10.28}$$

For $t = 1$, this is one of the announced inequalities.

Step 3: Taylor expansion of the function value. With the same start as in the proof of Proposition 10.53 and subtracting the term $\frac{1}{2}\langle s, \mathrm{Hess}f(x)[s]\rangle$ on both sides, we get

$$\begin{aligned} &f(\mathrm{Exp}_x(s)) - f(x) - \langle \mathrm{grad}f(x), s\rangle - \frac{1}{2}\langle s, \mathrm{Hess}f(x)[s]\rangle \\ &\qquad = \int_0^1 \big\langle P_{ts}^{-1}\mathrm{grad}f(\gamma(t)) - \mathrm{grad}f(x) - \mathrm{Hess}f(x)[ts], s\big\rangle\,\mathrm{d}t. \end{aligned}$$

Using (10.28) and Cauchy–Schwarz, it follows that

$$\begin{aligned} &\left|f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \mathrm{grad}f(x)\rangle - \frac{1}{2}\langle s, \mathrm{Hess}f(x)[s]\rangle\right| \\ &\qquad \leq \int_0^1 t^2\frac{L}{2}\|s\|^3\mathrm{d}t = \frac{L}{6}\|s\|^3, \end{aligned}$$

as announced. □

The following corollary shows that the regularity assumption A6.7 (p135) holds for the exponential retraction over its whole domain provided $\mathrm{Hess}f$ is L-Lipschitz (Definition 10.49). See also Exercise 10.87.

Corollary 10.56. *If $f\colon \mathcal{M} \to \mathbb{R}$ has L-Lipschitz continuous Hessian, then*

$$\left|f(\mathrm{Exp}_x(s)) - f(x) - \langle s, \mathrm{grad}f(x)\rangle - \frac{1}{2}\langle s, \mathrm{Hess}f(x)[s]\rangle\right| \leq \frac{L}{6}\|s\|^3,$$

$$\textit{and } \big\|P_s^{-1}\mathrm{grad}f(\mathrm{Exp}_x(s)) - \mathrm{grad}f(x) - \mathrm{Hess}f(x)[s]\big\| \leq \frac{L}{2}\|s\|^2,$$

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for all (x, s) in the domain of the exponential map.

To close this section, the following statement from [LKB22b] provides Lipschitz-type bounds for pullbacks through arbitrary retractions so long as we restrict our attention to a compact subset of the tangent bundle. The proof (omitted) merely uses the fact that if $f \circ \mathrm{R}$ is sufficiently many times continuously differentiable then its derivatives are bounded on any compact set. In turn, that can be used to bound truncation errors on Taylor expansions of $f \circ \mathrm{R}_x$ in $\mathrm{T}_x\mathcal{M}$ uniformly in x over a compact set. This is convenient to verify typical regularity assumptions such as A4.3, A6.6 and A6.7, though it should be noted that the constants L_1, L_2 below exist merely owing to the compactness-and-continuity argument: this provides little insight (let alone control) over those constants.

Lemma 10.57. *Consider a retraction R on $\mathcal{M}$, a compact subset $\mathcal{K} \subseteq \mathcal{M}$ and a continuous, nonnegative function $r\colon \mathcal{K} \to \mathbb{R}$. The set*

$$\mathcal{T} = \{(x, s) \in \mathrm{T}\mathcal{M} : x \in \mathcal{K} \text{ and } \|s\| \leq r(x)\}$$

is compact in the tangent bundle $\mathrm{T}\mathcal{M}$ (Exercise 10.31). Assume $f\colon \mathcal{M} \to \mathbb{R}$ is twice continuously differentiable. There exists a constant L_1 such that, for all $(x, s) \in \mathcal{T}$, with $\hat{f}_x = f \circ \mathrm{R}_x$, we have

$$|f(\mathrm{R}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle| \leq \frac{L_1}{2}\|s\|^2,$$
$$\left\|\mathrm{grad}\hat{f}_x(s) - \mathrm{grad}\hat{f}_x(0)\right\| \leq L_1\|s\|,$$

and $\|\mathrm{Hess}\hat{f}_x(0)\| \leq L_1$ for all $x \in \mathcal{K}$. If additionally f is three times continuously differentiable, then there exists a constant L_2 such that, for all $(x, s) \in \mathcal{T}$,

$$\left|f(\mathrm{R}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle - \frac{1}{2}\langle s, \mathrm{Hess}\hat{f}_x(0)[s]\rangle\right| \leq \frac{L_2}{6}\|s\|^3,$$
$$\left\|\mathrm{grad}\hat{f}_x(s) - \mathrm{grad}\hat{f}_x(0) - \mathrm{Hess}\hat{f}_x(0)[s]\right\| \leq \frac{L_2}{2}\|s\|^2,$$
$$\left\|\mathrm{Hess}\hat{f}_x(s) - \mathrm{Hess}\hat{f}_x(0)\right\| \leq L_2\|s\|.$$

(Recall $\mathrm{grad}\hat{f}_x(0) = \mathrm{grad} f(x)$ and, if the retraction is second order, $\mathrm{Hess}\hat{f}_x(0) = \mathrm{Hess} f(x)$.)

The exercise below has clear implications for the regularity assumptions A4.3 (p59) and A6.6 (p135). See also Exercise 10.87.

Exercise 10.58. *Let $f\colon \mathcal{M} \to \mathbb{R}$ be twice continuously differentiable on a manifold $\mathcal{M}$ equipped with a retraction R. Assume we have*

$$|f(\mathrm{R}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle| \leq \frac{L}{2}\|s\|^2 \tag{10.29}$$

for all (x, s) in a neighborhood of the zero section in the tangent bundle. With $\hat{f}_x = f \circ \mathrm{R}_x$, show that $\|\mathrm{Hess}\hat{f}_x(0)\| \leq L$. Deduce that if R is second order then the

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inequalities (10.29) *hold only if* $\mathrm{grad} f$ *is* L*-Lipschitz continuous. With* $\mathrm{R} = \mathrm{Exp}$ *in particular, verify that the three following claims are equivalent:*

1. *Inequalities* (10.29) *hold in a neighborhood of the zero section in* $\mathrm{T}\mathcal{M}$*;*
2. $\mathrm{grad} f$ *is* L*-Lipschitz continuous;*
3. *Inequalities* (10.29) *hold over the whole domain of* Exp.

Exercise 10.59. *Give a proof of Proposition 10.50, for example by adapting that of Proposition 10.45.*

10.5 Transporters

The strong properties of parallel transports (Section 10.3) make them great for theoretical purposes, and in some cases they can even be computed via explicit expressions. In general though, computing parallel transports involves numerically solving ordinary differential equations, which is typically too expensive in practice. Furthermore, we may want to dispense with the need to choose a curve connecting x and y explicitly to transport vectors from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_y\mathcal{M}$, as this may add to the computational burden (e.g., require computing $\mathrm{Log}_x(y)$ if we mean to transport along minimizing geodesics).

As an alternative, we define a poor man's version of parallel transports called *transporters*.[5] There is no need for a Riemannian structure or connection. Informally, for x and y close enough to one another, we aim to define linear maps of the form

$$\mathrm{T}_{y \leftarrow x} \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{M},$$

with $\mathrm{T}_{x \leftarrow x}$ in particular being the identity map. If $\mathcal{M}$ is an embedded submanifold of a Euclidean space, we present a simple transporter based on orthogonal projections to tangent spaces in Proposition 10.66.

It is natural and convenient to ask that these maps vary smoothly with respect to x and y. One indirect way to make sense of this statement would be to require that the map $((x, u), y) \mapsto \mathrm{T}_{y \leftarrow x} u$ be smooth from (an open submanifold of) $\mathrm{T}\mathcal{M} \times \mathcal{M}$ to $\mathrm{T}\mathcal{M}$. However, it is more instructive (and eventually more comfortable) to endow the set of linear maps between tangent spaces of two manifolds with a smooth structure. (Here, the two manifolds are the same.) Once this is done, we can formalize the notion of smoothness for a map $(x, y) \mapsto \mathrm{T}_{y \leftarrow x}$. This is in direct analogy with how we defined the tangent bundle $\mathrm{T}\mathcal{M}$ as a disjoint union of tangent spaces, associating a linear space $\mathrm{T}_x\mathcal{M}$ to each point $x \in \mathcal{M}$. Here, we associate to each pair $(x, y) \in \mathcal{M} \times \mathcal{N}$ the linear space of linear maps from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_y\mathcal{N}$. The proof is an exercise.

[5] This is different from the notion of *vector transport* as defined in [AMS08, §8.1]: we connect both concepts at the end of this section.

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Proposition 10.60. *For manifolds $\mathcal{M}$ and $\mathcal{N}$ of dimensions m and n, the disjoint union of linear maps from the tangent spaces of $\mathcal{M}$ to those of $\mathcal{N}$,*

$$\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{N}) = \{(x, y, \mathcal{L}) \,|\, x \in \mathcal{M}, y \in \mathcal{N} \text{ and } \mathcal{L} \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{N} \text{ is linear}\},$$

is itself a manifold, with charts as follows. For any pair of charts $(\mathcal{U}, \varphi)$ and $(\mathcal{V}, \psi)$ of $\mathcal{M}$ and $\mathcal{N}$, respectively, pick local frames on $\mathcal{U}$ and $\mathcal{V}$ as in Proposition 8.51; then,

$$\Phi(x, y, \mathcal{L}) = (\varphi(x), \psi(y), \mathrm{Mat}(\mathcal{L})) \in \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R}^{n \times m}$$

is a chart on $\pi^{-1}(\mathcal{U} \times \mathcal{V})$, where $\mathrm{Mat}(\mathcal{L})$ is the matrix that represents $\mathcal{L}$ with respect to the bases of $\mathrm{T}_x\mathcal{M}$ and $\mathrm{T}_y\mathcal{N}$ provided by the local frames, and $\pi(x, y, \mathcal{L}) = (x, y)$ is the projector from $\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{N})$ to $\mathcal{M} \times \mathcal{N}$.

The manifold $\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{N})$ is a *vector bundle* of $\mathcal{M} \times \mathcal{N}$ in that it (smoothly) attaches a linear space to each point of that manifold. Maps such as transporters defined below have the property that they map (x, y) to $(x, y, \mathcal{L})$ for some $\mathcal{L}$: these are called *sections* of the vector bundle. In the same way, vector fields are called sections of the tangent bundle.

Definition 10.61. *Given a manifold $\mathcal{M}$, let $\mathcal{V}$ be open in $\mathcal{M} \times \mathcal{M}$ such that $(x, x) \in \mathcal{V}$ for all $x \in \mathcal{M}$. A* transporter *on $\mathcal{V}$ is a smooth map*

$$\mathrm{T} \colon \mathcal{V} \to \mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M}) \colon (x, y) \mapsto \mathrm{T}_{y \leftarrow x}$$

such that $\mathrm{T}_{y \leftarrow x}$ is linear from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_y\mathcal{M}$ and $\mathrm{T}_{x \leftarrow x}$ is the identity.

In this definition, smoothness of T is understood with $\mathcal{V}$ as an open submanifold of the product manifold $\mathcal{M} \times \mathcal{M}$ and $\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M})$ equipped with the smooth structure of Proposition 10.60. Formally, this means that for any pair $(\bar{x}, \bar{y}) \in \mathcal{V}$ and local frames defined on neighborhoods $\mathcal{U}_{\bar{x}}$ and $\mathcal{U}_{\bar{y}}$, the matrix that represents $\mathrm{T}_{y \leftarrow x}$ with respect to these local frames varies smoothly with (x, y) in $\mathcal{U}_{\bar{x}} \times \mathcal{U}_{\bar{y}}$. We detail this in the proof of the next proposition, which shows that inverting the linear maps of a transporter yields a transporter.

Proposition 10.62. *For a transporter T on $\mathcal{V}$, let $\mathcal{V}'$ be the set of pairs $(x, y) \in \mathcal{V}$ such that $\mathrm{T}_{x \leftarrow y}$ is invertible. Then, the maps*

$$\mathrm{T}'_{y \leftarrow x} = (\mathrm{T}_{x \leftarrow y})^{-1} \colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{M}$$

define a transporter T' on $\mathcal{V}'$.

Proof. For all $x \in \mathcal{M}$, since $\mathrm{T}_{x \leftarrow x}$ is the identity map, clearly $(x, x) \in \mathcal{V}'$ and $\mathrm{T}'_{x \leftarrow x}$ is itself the identity. Likewise, for all $(x, y) \in \mathcal{V}'$, it is clear that $\mathrm{T}'_{y \leftarrow x}$ is linear. It remains to argue that $\mathcal{V}'$ is open in $\mathcal{M} \times \mathcal{M}$ and that T' is smooth from $\mathcal{V}'$ to $\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M})$.

To this end, consider an arbitrary pair $(\bar{x}, \bar{y}) \in \mathcal{V}'$ and let $U_1, \ldots, U_d$ be a local frame on a neighborhood $\mathcal{U}_{\bar{x}}$ of $\bar{x}$ with $d = \dim \mathcal{M}$—see Proposition 8.51. Likewise, let $W_1, \ldots, W_d$ be a local frame on a neighborhood $\mathcal{U}_{\bar{y}}$ of $\bar{y}$. If need be,

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reduce $\mathcal{U}_{\bar{x}}$ and $\mathcal{U}_{\bar{y}}$ to smaller neighborhoods of $\bar{x}$ and $\bar{y}$ so that $\mathcal{U}_{\bar{x}} \times \mathcal{U}_{\bar{y}} \subseteq \mathcal{V}$. (We can do this because $\mathcal{V}$ is open in $\mathcal{M} \times \mathcal{M}$, hence it is a union of products of open sets in $\mathcal{M}$. One of these products, say $\tilde{\mathcal{U}} \times \hat{\mathcal{U}}$, contains $(\bar{x}, \bar{y})$, as otherwise it would not be in $\mathcal{V}$; thus: $\bar{x} \in \tilde{\mathcal{U}}$ and $\bar{y} \in \hat{\mathcal{U}}$. Replace $\mathcal{U}_{\bar{x}}$ by its intersection with $\tilde{\mathcal{U}}$, and similarly for $\mathcal{U}_{\bar{y}}$: now $\mathcal{U}_{\bar{x}} \times \mathcal{U}_{\bar{y}} \subseteq \mathcal{V}$ is a neighborhood of $(\bar{x}, \bar{y})$ and the local frames are well defined.) Since T is smooth, the matrix $G(x, y)$ in $\mathbb{R}^{d \times d}$ that represents $\mathrm{T}_{x \leftarrow y}$ with respect to the bases $U_1(x), \ldots, U_d(x)$ and $W_1(y), \ldots, W_d(y)$ varies smoothly with (x, y) in $\mathcal{U}_{\bar{x}} \times \mathcal{U}_{\bar{y}}$. In particular, the function $(x, y) \mapsto \det G(x, y)$ is smooth on this domain, so that the subset of $\mathcal{U}_{\bar{x}} \times \mathcal{U}_{\bar{y}}$ over which $\det G(x, y) \neq 0$ (that is, over which $\mathrm{T}_{x \leftarrow y}$ is invertible) is open, and it contains $(\bar{x}, \bar{y})$. In other words: this subset is a neighborhood of $(\bar{x}, \bar{y})$ in $\mathcal{V}'$. Since each point in $\mathcal{V}'$ admits such a neighborhood, we find that $\mathcal{V}'$ is open. Furthermore, the matrix that represents $\mathrm{T}'_{y \leftarrow x}$ is simply $G(x, y)^{-1}$. This is a smooth function of (x, y) on the open set where the inverse is well defined, confirming that T' is smooth from $\mathcal{V}'$ to $\mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M})$. □

With similar developments, we also get the following result once we equip the manifold with a Riemannian metric.

Proposition 10.63. *Let $\mathcal{M}$ be a Riemannian manifold and let* T *be a transporter for $\mathcal{M}$ on $\mathcal{V}$. Then,* T′ *defined by the maps*

$$\mathrm{T}'_{y \leftarrow x} = (\mathrm{T}_{x \leftarrow y})^* : \mathrm{T}_x\mathcal{M} \to \mathrm{T}_y\mathcal{M}$$

is a transporter on $\mathcal{V}$. (As always, the superscript $*$ denotes the adjoint, here with respect to the Riemannian metric at x and y.)

Proof sketch. Compared to Proposition 10.62 (and using the same notation), an extra step in the proof is to show that, using local frames, the Riemannian metric can be represented as a smooth map from x to $M(x)$: a symmetric, positive definite matrix of size d which allows us to write $\langle u, v \rangle_x = \bar{u}^\top M(x) \bar{v}$ with $\bar{u}, \bar{v} \in \mathbb{R}^d$ denoting the coordinate vectors of u, v in the same local frame. Upon doing so, it is straightforward to show that the matrix which represents $(\mathrm{T}_{x \leftarrow y})^*$ is $M(y)^{-1} G(x, y)^\top M(x)$, which is indeed smooth in (x, y). □

Upon choosing a smoothly varying collection of curves that uniquely connect pairs of nearby points on $\mathcal{M}$, it is easy to construct a transporter from parallel transport along those curves (with respect to some connection). One way of choosing such families of curves is via a retraction.

Conveniently, the differentials of a retraction also provide a transporter. This is because, with $y = \mathrm{R}_x(v)$, by perturbing v in $\mathrm{T}_x\mathcal{M}$ we perturb $\mathrm{R}_x(v)$ away from y, thus producing a tangent vector in $\mathrm{T}_y\mathcal{M}$. That is a good alternative when parallel transports are out of reach.

Proposition 10.64. *For a retraction* R *on a manifold $\mathcal{M}$, let $\mathcal{T}$ be a neighborhood of the zero section of $\mathrm{T}\mathcal{M}$ such that $E(x, v) = (x, \mathrm{R}_x(v))$ is a diffeomorphism from $\mathcal{T}$ to $\mathcal{V} = E(\mathcal{T})$—such neighborhoods exist by Corollary 10.27. For*

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our purpose, this means $(x, y) \mapsto (x, \mathrm{R}_x^{-1}(y))$ *is a diffeomorphism from* $\mathcal{V}$ *to* $\mathcal{T}$*, yielding a smooth choice of curves joining pairs* (x, y)*.*

1. *Assume* $\mathcal{T}_x = \{v \in \mathrm{T}_x\mathcal{M} : (x, v) \in \mathcal{T}\}$ *is star-shaped around the origin for all* x*. Parallel transport along retraction curves defines a transporter on* $\mathcal{V}$ *via* $\mathrm{T}_{y \leftarrow x} = \mathrm{PT}^c_{1 \leftarrow 0}$*, where* $c(t) = \mathrm{R}_x(tv)$ *and* $v = \mathrm{R}_x^{-1}(y)$*.*
2. *The differentials of the retraction define a transporter on* $\mathcal{V}$ *via* $\mathrm{T}_{y \leftarrow x} = \mathrm{DR}_x(v)$*, where* $v = \mathrm{R}_x^{-1}(y)$*.*

Proof sketch. The domain $\mathcal{V} \subseteq \mathcal{M} \times \mathcal{M}$ is open and indeed contains all pairs (x, x) since $E(x, 0) = (x, x)$. For both proposed transporters, it is clear that $\mathrm{T}_{x \leftarrow x}$ is the identity and that $\mathrm{T}_{y \leftarrow x}$ is a linear map from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_y\mathcal{M}$. Smoothness for parallel transport can be argued with tools from ordinary differential equations (it takes some work). Smoothness for the retraction-based transport follows by composition of smooth maps since $\mathrm{T}_{y \leftarrow x} = \mathrm{DR}_x(\mathrm{R}_x^{-1}(y))$. □

Example 10.65. *Transporters can be used to transport linear maps between certain tangent spaces to other tangent spaces. This is useful notably in defining a Riemannian version of the famous BFGS algorithm. For example, if* $\mathcal{A}$ *is a linear map from* $\mathrm{T}_x\mathcal{M}$ *to* $\mathrm{T}_x\mathcal{M}$*, then we may transport it to a linear map from* $\mathrm{T}_y\mathcal{M}$ *to* $\mathrm{T}_y\mathcal{M}$ *in at least three ways using a transporter* T*:*

$$\mathrm{T}_{y \leftarrow x} \circ \mathcal{A} \circ \mathrm{T}_{x \leftarrow y}, \qquad (\mathrm{T}_{x \leftarrow y})^* \circ \mathcal{A} \circ \mathrm{T}_{x \leftarrow y}, \qquad (\mathrm{T}_{x \leftarrow y})^{-1} \circ \mathcal{A} \circ \mathrm{T}_{x \leftarrow y}.$$

If $\mathcal{A}$ *is self-adjoint, then so is the second operator. If the transporter is obtained through parallel transport as in Proposition 10.64 and the curve connecting* x *to* y *is the same as the curve connecting* y *to* x *(for example, if we use unique minimizing geodesics), then all three operators are equal: see Proposition 10.36.*

For manifolds embedded in Euclidean spaces, an especially convenient transporter is given by orthogonal projectors to the tangent spaces. In contrast to Proposition 10.64, it does not involve retractions.

Proposition 10.66. *Let* $\mathcal{M}$ *be an embedded submanifold of a Euclidean space* $\mathcal{E}$*. For all* $x, y \in \mathcal{M}$*, exploiting the fact that both* $\mathrm{T}_x\mathcal{M}$ *and* $\mathrm{T}_y\mathcal{M}$ *are subspaces of* $\mathcal{E}$*, define the linear maps*

$$\mathrm{T}_{y \leftarrow x} = \mathrm{Proj}_y\big|_{\mathrm{T}_x\mathcal{M}},$$

where Proj_y *is the orthogonal projector from* $\mathcal{E}$ *to* $\mathrm{T}_y\mathcal{M}$*, here restricted to* $\mathrm{T}_x\mathcal{M}$*. This is a transporter on all of* $\mathcal{M} \times \mathcal{M}$*.*

Proof. By design, $\mathrm{T}_{x \leftarrow x}$ is the identity and $\mathrm{T}_{y \leftarrow x}$ is linear from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_y\mathcal{M}$. Moreover, T is smooth as can be deduced by an argument along the same lines as in Exercise 3.66. □

For a quotient manifold $\mathcal{M} = \overline{\mathcal{M}}/\sim$, in the same way that we discussed conditions for a retraction $\overline{\mathrm{R}}$ on the total space $\overline{\mathcal{M}}$ to induce a retraction R on the quotient manifold $\mathcal{M}$, it is tempting to derive a transporter T on $\mathcal{M}$ from a transporter $\overline{\mathrm{T}}$ on $\overline{\mathcal{M}}$. We show through an example how this can be done.

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Example 10.67. *Consider the Grassmann manifold* $\mathrm{Gr}(n,p) = \mathrm{St}(n,p)/\sim$ *(Section 9.16). Equip the total space with the polar retraction* (7.24), $\overline{\mathrm{R}}_X(V) = \mathrm{pfactor}(X+V)$, *and with the projection transporter,*

$$\overline{\mathrm{T}}_{Y\leftarrow X} = \mathrm{Proj}_Y^{\mathrm{St}}|_{\mathrm{T}_X\mathrm{St}(n,p)},$$

the orthogonal projector from $\mathbb{R}^{n\times p}$ *to* $\mathrm{T}_Y\mathrm{St}(n,p)$ *restricted to* $\mathrm{T}_X\mathrm{St}(n,p)$. *This transporter is defined globally on* $\mathrm{St}(n,p)\times\mathrm{St}(n,p)$. *Our tentative transporter on* $\mathrm{Gr}(n,p)$ *is:*

$$\mathrm{T}_{[Y]\leftarrow[X]}(\xi) = \mathrm{D}\pi(Y)\left[\overline{\mathrm{T}}_{Y\leftarrow X}(\mathrm{lift}_X(\xi))\right], \tag{10.30}$$

where X *is an arbitrary representative of* $[X]$, *and* Y *is a representative of* $[Y]$ *such that* $Y = \overline{\mathrm{R}}_X(V)$ *for some* $V \in \mathrm{H}_X$, *assuming one exists. When such a choice of* Y *and* V *exists, it is unique. Indeed, consider the map*

$$\begin{aligned} E\colon \mathrm{TGr}(n,p) &\to \mathrm{Gr}(n,p)\times\mathrm{Gr}(n,p) \\ \colon ([X],\xi) &\mapsto E([X],\xi) = ([X],[\overline{\mathrm{R}}_X(V)]), \end{aligned} \tag{10.31}$$

where $V = \mathrm{lift}_X(\xi)$. *In Exercise 10.71, we find that* E *from* $\mathrm{TGr}(n,p)$ *to* $\mathcal{V} = E(\mathrm{TGr}(n,p))$ *is smoothly invertible. In other words: if* $[Y]$ *can be reached from* $[X]$ *through retraction, it is so by a unique tangent vector* ξ; *the latter has a specific horizontal lift* V *once we choose a specific representative* X. *Furthermore, since* E^{-1} *is continuous,* $\mathcal{V}$ *is open. Finally,* $\mathcal{V}$ *contains all pairs of the form* $([X],[X])$. *This set* $\mathcal{V}$ *is meant to be the domain of* T.

Now restricting our discussion to $\mathcal{V}$, *we rewrite* (10.30) *equivalently as*

$$\begin{aligned} \mathrm{lift}_Y\big(\mathrm{T}_{[Y]\leftarrow[X]}(\xi)\big) &= \mathrm{Proj}_Y^{\mathrm{H}}\left(\overline{\mathrm{T}}_{Y\leftarrow X}(\mathrm{lift}_X(\xi))\right) \\ &= \mathrm{Proj}_Y^{\mathrm{H}}\left(\mathrm{lift}_X(\xi)\right), \end{aligned} \tag{10.32}$$

where we used that $\mathrm{Proj}_Y^{\mathrm{H}}\circ\mathrm{Proj}_Y^{\mathrm{St}} = \mathrm{Proj}_Y^{\mathrm{H}}$. *We must check (a) that* $\mathrm{T}_{[Y]\leftarrow[X]}$ *is well defined, and (b) that it defines a transporter, both on* $\mathcal{V}$.

For (a), we must check that the right-hand side of (10.30) *does not depend on our choice of representatives* X *and* Y. *To this end, consider* (10.32). *Recall from Example 9.26 that if we choose the representative* XQ *instead of* X *for* $[X]$ *with some arbitrary* $Q \in \mathrm{O}(p)$, *then* $\mathrm{lift}_{XQ}(\xi) = \mathrm{lift}_X(\xi)Q$. *The representative* Y *also changes as a result. Indeed, given* $V \in \mathrm{H}_X$ *such that* $\overline{\mathrm{R}}_X(V) = Y$, *we know that* $VQ \in \mathrm{H}_{XQ}$ *is such that* $\overline{\mathrm{R}}_{XQ}(VQ) = YQ$ *(this is specific to the polar retraction by* (9.9)*), and this is the only horizontal vector at* XQ *that maps to* $[Y]$. *Since* $\mathrm{Proj}_Y^{\mathrm{H}} = I_n - YY^\top = \mathrm{Proj}_{YQ}^{\mathrm{H}}$, *we find that*

$$\begin{aligned} \mathrm{lift}_{YQ}\big(\mathrm{T}_{[Y]\leftarrow[X]}(\xi)\big) &= \mathrm{Proj}_{YQ}^{\mathrm{H}}\left(\mathrm{lift}_{XQ}(\xi)\right) \\ &= (I_n - YY^\top)\,\mathrm{lift}_X(\xi)Q \\ &= \mathrm{lift}_Y\big(\mathrm{T}_{[Y]\leftarrow[X]}(\xi)\big)\,Q. \end{aligned}$$

This confirms that the lifted vectors correspond to each other in the appropriate way, that is, the result $\mathrm{T}_{[Y]\leftarrow[X]}(\xi)$ *does not depend on our choice of representative* X.

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Regarding (b), it is clear that $\mathrm{T}_{[Y]\leftarrow[X]}$ *is a linear map from* $\mathrm{T}_{[X]}\mathrm{Gr}(n,p)$ *to* $\mathrm{T}_{[Y]}\mathrm{Gr}(n,p)$*, as a composition of linear maps. Likewise,* T *is smooth as a composition of smooth maps (this also follows from Exercise 10.71, which shows that given* $([X],[Y]) \in \mathcal{V}$*, for any choice of representative* X*, there is a smooth choice of* Y *and* V *(horizontal) such that* $Y = \overline{\mathrm{R}}_X(V)$*). It is easy to see that* $\mathrm{T}_{[X]\leftarrow[X]}$ *is the identity. Finally, we already checked that* $\mathcal{V}$ *is an appropriate domain for a transporter.*

How do we use this transporter in practice? If we are simply given two representatives X *and* Y *and the lift* U *of* ξ *at* X*, then before applying (10.32) we must replace* Y *by* YQ*, for the unique* Q *such that there exists* $V \in \mathrm{H}_X$ *with* $\overline{\mathrm{R}}_X(V) = YQ$*. This can be done if and only if* $X^\top Y$ *is invertible. Explicitly, one can reason from Exercise 10.71 that* Q *is nothing but the polar factor of* $X^\top Y$*. Then, we can follow this procedure:*

1. *Compute* $Q \in \mathrm{O}(p)$ *via SVD, as* $Q = \tilde{U}\tilde{V}^\top$ *with* $\tilde{U}\tilde{\Sigma}\tilde{V}^\top = X^\top Y$*;*
2. *By (10.32),* $\mathrm{lift}_{YQ}\big(\mathrm{T}_{[Y]\leftarrow[X]}(\xi)\big) = \mathrm{Proj}^{\mathrm{H}}_{YQ}(\mathrm{lift}_X(\xi)) = U - Y(Y^\top U)$*;*
3. *Finally,* $\mathrm{lift}_Y\big(\mathrm{T}_{[Y]\leftarrow[X]}(\xi)\big) = (U - Y(Y^\top U))Q^\top$*.*

Often times though, Y *is a point that was generated by retraction of some horizontal vector from* X*. If that retraction is the polar retraction, then using this transporter is straightforward:* $X^\top Y$ *is symmetric and positive definite, hence its polar factor is* $Q = I_p$*, and it is sufficient to compute* $U - Y(Y^\top U)$*.*

In closing, we connect the notion of transporter (used in [HGA15, §4.3]) to that of vector transport (favored in [AMS08, Def. 8.1.1]).

Definition 10.68. *A* vector transport *on a manifold* $\mathcal{M}$ *is a smooth map*

$$(x,u,v) \mapsto \mathrm{VT}_{(x,u)}(v)$$

from the Whitney sum *(which can be endowed with a smooth structure)*

$$\mathrm{T}\mathcal{M} \oplus \mathrm{T}\mathcal{M} = \{(x,u,v) : x \in \mathcal{M} \text{ and } u,v \in \mathrm{T}_x\mathcal{M}\}$$

to $\mathrm{T}\mathcal{M}$*, satisfying the following for some retraction* R *on* $\mathcal{M}$*:*

1. $\mathrm{VT}_{(x,u)}$ *is a linear map from the tangent space at* x *to the tangent space at* $\mathrm{R}_x(u)$ *for all* $(x,u) \in \mathrm{T}\mathcal{M}$*; and*
2. $\mathrm{VT}_{(x,0)}$ *is the identity on* $\mathrm{T}_x\mathcal{M}$ *for all* $x \in \mathcal{M}$*.*

Equivalently, we can define a vector transport associated to a retraction R as a smooth map $\mathrm{VT}\colon \mathrm{T}\mathcal{M} \to \mathcal{L}(\mathrm{T}\mathcal{M}, \mathrm{T}\mathcal{M})$ such that $\mathrm{VT}_{(x,u)}$ is a linear map from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_{\mathrm{R}_x(u)}\mathcal{M}$ and $\mathrm{VT}_{(x,0)}$ is the identity on $\mathrm{T}_x\mathcal{M}$. From this perspective, it is clear that a transporter T and a retraction R can be combined to define a vector transport through $\mathrm{VT}_{(x,u)} = \mathrm{T}_{\mathrm{R}_x(u)\leftarrow x}$. However, not all vector transports are of this form because in general we could have $\mathrm{VT}_{(x,u)} \neq \mathrm{VT}_{(x,w)}$ even if $\mathrm{R}_x(u) = \mathrm{R}_x(w)$, which the transporter construction does not allow. The other way around, a vector transport with associated retraction R can be used to define

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a transporter if we first restrict the domain such that $(x, u) \mapsto (x, \mathrm{R}_x(u))$ admits a smooth inverse (see Corollary 10.27).

Exercise 10.69. *Give a proof of Proposition 10.60.*

Exercise 10.70. *Give a proof of Proposition 10.63.*

Exercise 10.71. *With notation as in Example 10.67, show that E (10.31) is invertible. Furthermore, show that given two arbitrary representatives X and Y of $[X]$ and $[\overline{\mathrm{R}}_X(V)]$ (respectively), V is given by*

$$V = Y(X^\top Y)^{-1} - X, \tag{10.33}$$

and deduce that the inverse of E is smooth. From this formula, it is also apparent that $[Y]$ can be reached from $[X]$ if and only if $X^\top Y$ is invertible. Compare with Exercise 7.2.

Exercise 10.72. *For the orthogonal group ($\mathcal{M} = \mathrm{O}(n)$) or the group of rotations ($\mathcal{M} = \mathrm{SO}(n)$) as a Riemannian submanifold of $\mathbb{R}^{n\times n}$ (see Section 7.4), it is natural to consider the following transporter:*

$$\mathrm{T}_{Y \leftarrow X}(U) = YX^\top U, \tag{10.34}$$

where $X, Y \in \mathcal{M}$ are orthogonal matrices of size n and $U \in \mathrm{T}_X\mathcal{M}$ is such that $X^\top U$ is skew-symmetric. Show that this is indeed a transporter and that it is isometric. Then, show that this is not *parallel transport along geodesics (see Exercise 7.3). If we represent tangent vectors $U = X\Omega$ simply as their skew-symmetric part Ω, then this transporter requires no computations.*

Exercise 10.73. *Let R be a retraction on a Riemannian manifold $\mathcal{M}$. The differentiated retraction plays a special role as a link between the Riemannian gradient and Hessian of $f\colon \mathcal{M} \to \mathbb{R}$ and the (classical) gradients and Hessians of the pullbacks $\hat{f} = f \circ \mathrm{R}_x\colon \mathrm{T}_x\mathcal{M} \to \mathbb{R}$.*

Prove the following identities [ABBC20, §6]: if f is differentiable, then

$$\operatorname{grad}\hat{f}(s) = T_s^* \operatorname{grad} f(\mathrm{R}_x(s)), \tag{10.35}$$

where $T_s = \mathrm{D}\mathrm{R}_x(s)$ is a linear map from $\mathrm{T}_x\mathcal{M}$ to $\mathrm{T}_{\mathrm{R}_x(s)}\mathcal{M}$, and T_s^* is its adjoint. If f is twice differentiable, then

$$\operatorname{Hess}\hat{f}(s) = T_s^* \circ \operatorname{Hess} f(\mathrm{R}_x(s)) \circ T_s + W_s, \tag{10.36}$$

with W_s a self-adjoint linear map on $\mathrm{T}_x\mathcal{M}$ defined by

$$\langle \dot{s}, W_s(\dot{s}) \rangle_x = \langle \operatorname{grad} f(\mathrm{R}_x(s)), c''(0) \rangle_{\mathrm{R}_x(s)}, \tag{10.37}$$

where $c''(0) = \frac{\mathrm{D}}{\mathrm{d}t} c'(0)$ is the initial intrinsic acceleration of the smooth curve $c(t) = \mathrm{R}_x(s + t\dot{s})$. Argue that W_s is indeed linear and self-adjoint.

Check that these formulas generalize Propositions 8.59 and 8.71 (as well as their embedded counter-parts, Propositions 3.59 and 5.45).

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As a comment: For all $u, v \in \mathrm{T}_x\mathcal{M}$, we can use (10.37) to compute $\langle u, W_s(v)\rangle_x$ owing to the polarization identity

$$\langle u, W_s(v)\rangle_x = \frac{1}{4}\big(\langle u+v, W_s(u+v)\rangle_x - \langle u-v, W_s(u-v)\rangle_x\big).$$

This is why (10.37) fully determines W_s.

10.6 Finite difference approximation of the Hessian

In order to minimize a smooth function $f\colon \mathcal{M} \to \mathbb{R}$ on a Riemannian manifold, several optimization algorithms (notably those in Chapter 6) require computation of the Riemannian Hessian applied to a vector: $\mathrm{Hess} f(x)[u]$. Since obtaining an explicit expression for the Hessian may be tedious,[6] it is natural to explore avenues to approximate it numerically. To this end, we consider *finite difference approximations*.

For any smooth curve $c\colon I \to \mathcal{M}$ such that $c(0) = x$ and $c'(0) = u$, it holds that

$$\mathrm{Hess} f(x)[u] = \nabla_u \mathrm{grad} f = \frac{\mathrm{D}}{\mathrm{d}t}(\mathrm{grad} f \circ c)(0). \tag{10.38}$$

Using Proposition 10.37, we can further rewrite the right-hand side in terms of parallel transport along c:

$$\mathrm{Hess} f(x)[u] = \lim_{t\to 0} \frac{\mathrm{PT}^c_{0\leftarrow t}(\mathrm{grad} f(c(t))) - \mathrm{grad} f(x)}{t}. \tag{10.39}$$

This suggests the approximation

$$\mathrm{Hess} f(x)[u] \approx \frac{\mathrm{PT}^c_{0\leftarrow \bar{t}}(\mathrm{grad} f(c(\bar{t}))) - \mathrm{grad} f(x)}{\bar{t}} \tag{10.40}$$

for some well-chosen $\bar{t} > 0$: small enough to be close to the limit (see Corollary 10.56 to quantify this error), large enough to avoid numerical issues. Of course, we could also use higher-order finite differences.

In light of Section 10.5, we may ask: is it legitimate to replace the parallel transport in (10.40) with a transporter? We already verified this for a special case in Example 5.32, where we considered a Riemannian submanifold $\mathcal{M}$ of a Euclidean space $\mathcal{E}$ with the transporter obtained by orthogonal projection to tangent spaces. In this section, we consider the general setting.

With a transporter T on a Riemannian manifold $\mathcal{M}$, we contemplate the following candidate approximation for the Hessian:

$$\mathrm{Hess} f(x)[u] \approx \frac{\mathrm{T}_{x\leftarrow c(\bar{t})}(\mathrm{grad} f(c(\bar{t}))) - \mathrm{grad} f(x)}{\bar{t}}. \tag{10.41}$$

Implementing this formula takes little effort compared to the hassle of deriving formulas for the Hessian by hand. For example, in the Manopt toolbox,

[6] See also Section 4.7 for a word regarding automatic differentiation.

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the default behavior when the Hessian is needed but unavailable is to fall back on (10.41) with $c(\bar{t}) = \mathrm{R}_x(\bar{t}u)$ and $\bar{t} > 0$ set such that $\|\bar{t}u\|_x = 2^{-14}$. This costs one retraction, one gradient evaluation (assuming $\operatorname{grad} f(x)$ is available), and one call to a transporter. Moreover, this satisfies radial linearity as required in A6.1 (p133). To justify (10.41), we generalize Proposition 10.37.

Proposition 10.74. *Let $c\colon I \to \mathcal{M}$ be a smooth curve on a Riemannian manifold equipped with a transporter T. For a fixed $t_0 \in I$, let $v_1, \ldots, v_d$ form a basis of $\mathrm{T}_{c(t_0)}\mathcal{M}$ and define the vector fields*

$$V_i(t) = \left(\mathrm{T}_{c(t_0)\leftarrow c(t)}\right)^{-1}(v_i). \tag{10.42}$$

Given a vector field $Z \in \mathfrak{X}(c)$, it holds that

$$\frac{\mathrm{D}}{\mathrm{d}t}Z(t_0) = \lim_{\delta\to 0} \frac{\mathrm{T}_{c(t_0)\leftarrow c(t_0+\delta)}Z(t_0+\delta) - Z(t_0)}{\delta} + \sum_{i=1}^{d} \alpha_i(t_0)\frac{\mathrm{D}}{\mathrm{d}t}V_i(t_0),$$

where $\alpha_1(t_0), \ldots, \alpha_d(t_0)$ are the coefficients of $Z(t_0)$ in the basis $v_1, \ldots, v_d$.

Proof. The vector fields $V_1, \ldots, V_d$ play a role similar to parallel frames. By Proposition 10.62, these vector fields depend smoothly on t in a neighborhood I_0 of t_0. Furthermore, I_0 can be chosen small enough so that $V_1(t), \ldots, V_d(t)$ form a basis of $\mathrm{T}_{c(t)}\mathcal{M}$ for each $t \in I_0$. Hence, there exists a unique set of smooth functions $\alpha_i\colon I_0 \to \mathbb{R}$ such that

$$Z(t) = \sum_{i=1}^{d} \alpha_i(t)V_i(t).$$

On the one hand, using properties of covariant derivatives, we see that

$$\frac{\mathrm{D}}{\mathrm{d}t}Z(t) = \sum_{i=1}^{d} \alpha_i'(t)V_i(t) + \alpha_i(t)\frac{\mathrm{D}}{\mathrm{d}t}V_i(t).$$

On the other hand, defining G as

$$G(t) = \mathrm{T}_{c(t_0)\leftarrow c(t)}(Z(t)) = \sum_{i=1}^{d} \alpha_i(t)v_i,$$

we find that

$$G'(t) = \sum_{i=1}^{d} \alpha_i'(t)v_i.$$

Combining both findings at t_0 using $V_i(t_0) = v_i$, it follows that

$$\frac{\mathrm{D}}{\mathrm{d}t}Z(t_0) = G'(t_0) + \sum_{i=1}^{d} \alpha_i(t_0)\frac{\mathrm{D}}{\mathrm{d}t}V_i(t_0).$$

Since G is a map between (open subsets of) linear spaces, we can write $G'(t_0)$ as a limit in the usual way. □

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Applying this to (10.38) yields a corollary relevant to formula (10.41).

Corollary 10.75. *For any smooth curve c on a Riemannian manifold $\mathcal{M}$ such that $c(0) = x$ and $c'(0) = u$, and for any transporter* T, orthonormal *basis $v_1, \ldots, v_d$ of $\mathrm{T}_x\mathcal{M}$ and associated vector fields $V_1, \ldots, V_d$ defined by*

$$V_i(t) = \left(\mathrm{T}_{x \leftarrow c(t)}\right)^{-1}(v_i), \tag{10.43}$$

it holds that

$$\mathrm{Hess} f(x)[u] = \lim_{t \to 0} \frac{\mathrm{T}_{x \leftarrow c(t)}(\mathrm{grad} f(c(t))) - \mathrm{grad} f(x)}{t} + \sum_{i=1}^{d} \langle \mathrm{grad} f(x), v_i \rangle_x \frac{\mathrm{D}}{\mathrm{d}t} V_i(0). \tag{10.44}$$

Thus we see that the approximation (10.41) is justified at or near a critical point. This is typically sufficient to obtain good performance with second-order optimization algorithms such as RTR.

The approximation is also justified at a general point x if the vectors $\frac{\mathrm{D}}{\mathrm{d}t} V_i(0)$ vanish. This is of course the case if we use parallel transport, recovering (10.39). Likewise, circling back to Example 5.32 for the case where $\mathcal{M}$ is a Riemannian submanifold of a Euclidean space $\mathcal{E}$ and the transporter is taken to be simply orthogonal projection to tangent spaces (Proposition 10.66), we also get the favorable simplification. As a reminder, this yields the particularly convenient formula

$$\mathrm{Hess} f(x)[u] = \lim_{t \to 0} \frac{\mathrm{Proj}_x(\mathrm{grad} f(c(t))) - \mathrm{grad} f(x)}{t}, \tag{10.45}$$

where Proj_x is the orthogonal projector from $\mathcal{E}$ to $\mathrm{T}_x\mathcal{M}$, $c(t)$ satisfies $c(0) = x$ and $c'(0) = u$, and $\mathrm{grad} f(c(t))$ is interpreted as a vector in $\mathcal{E}$.

10.7 Tensor fields and their covariant differentiation

Given a Riemannian manifold $\mathcal{M}$, we can think of a smooth vector field $U \in \mathfrak{X}(\mathcal{M})$ as a map from $\mathfrak{X}(\mathcal{M})$ to the set of smooth real-valued functions $\mathfrak{F}(\mathcal{M})$ as follows:

$$V \mapsto U(V) = \langle U, V \rangle .$$

This map is $\mathfrak{F}(\mathcal{M})$-linear in its argument, meaning

$$U(fV + gW) = fU(V) + gU(W)$$

for all $V, W \in \mathfrak{X}(\mathcal{M})$ and $f, g \in \mathfrak{F}(\mathcal{M})$. Likewise, we can think of the Riemannian metric itself as a map from $\mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M})$ to $\mathfrak{F}(\mathcal{M})$:

$$(U, V) \mapsto \langle U, V \rangle .$$

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This mapping is $\mathfrak{F}(\mathcal{M})$-linear in each of its arguments. These two maps are examples of *tensor fields* of order one and two, respectively.

Definition 10.76. *A* smooth tensor field T of order k *on a manifold* $\mathcal{M}$ *is a map*

$$T\colon \mathfrak{X}(\mathcal{M}) \times \cdots \times \mathfrak{X}(\mathcal{M}) \to \mathfrak{F}(\mathcal{M})$$

which is $\mathfrak{F}(\mathcal{M})$*-linear in each one of its* k *inputs. The set of such objects is denoted by* $\mathfrak{X}^k(\mathcal{M})$*. If the ordering of the inputs is irrelevant, we say* T *is a* symmetric (smooth) tensor field*. (See also Remark 10.84.)*

In our examples above, vector fields are identified with tensor fields of order one, while the Riemannian metric is a symmetric tensor field of order two. As a non-example, notice that the Riemannian connection ∇, conceived of as a map

$$\nabla\colon \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \times \mathfrak{X}(\mathcal{M}) \to \mathfrak{F}(\mathcal{M})\colon (U, V, W) \mapsto \langle \nabla_U V, W \rangle\,,$$

is *not* a tensor field, because it is only $\mathbb{R}$-linear in V, not $\mathfrak{F}(\mathcal{M})$-linear. Indeed, $\nabla_U(fV) = f\nabla_U V + (Uf)V$ for $f \in \mathfrak{F}(\mathcal{M})$.

Importantly, tensor fields are pointwise objects, in that they associate to each point $x \in \mathcal{M}$ a well-defined multilinear map (i.e., a *tensor*) on the tangent space $\mathrm{T}_x\mathcal{M}$—hence the name tensor *field*. In order to see this, consider a tensor field T of order k and a local frame $W_1, \ldots, W_d$ on a neighborhood $\mathcal{U}$ of x (Section 3.9). Then, the input vector fields $U_1, \ldots, U_k \in \mathfrak{X}(\mathcal{M})$ can each be expanded in the local frame as

$$U_i|_{\mathcal{U}} = f_{i,1}W_1 + \cdots + f_{i,d}W_d = \sum_{j=1}^{d} f_{i,j}W_j,$$

where the $f_{i,j}$ are smooth functions on $\mathcal{U}$. Working only on the domain $\mathcal{U}$ and using the linearity properties of tensor fields, we find[7]

$$T(U_1, \ldots, U_k) = \sum_{j_1=1}^{d} \cdots \sum_{j_k=1}^{d} f_{1,j_1} \cdots f_{k,j_k} T(W_{j_1}, \ldots, W_{j_k}).$$

Evaluating this function at $x' \in \mathcal{U}$, the result depends on $U_1, \ldots, U_k$ only through the values of the $f_{i,j}$ at x', that is, $T(U_1, \ldots, U_k)(x')$ depends on the vector fields only through $U_1(x'), \ldots, U_k(x')$. Moreover, the dependence is linear in each one.

This offers a useful perspective on tensor fields of order k: they associate to each point x of $\mathcal{M}$ a k-linear function on the tangent space at that point, namely,

$$T(x)\colon \mathrm{T}_x\mathcal{M} \times \cdots \times \mathrm{T}_x\mathcal{M} \to \mathbb{R}.$$

This function is defined by

$$T(x)(u_1, \ldots, u_k) = T(U_1, \ldots, U_k)(x),$$

[7] Via bump functions as in Section 5.6, we can extend the vector fields from $\mathcal{U}$ to all of $\mathcal{M}$, which is necessary to apply T. To be formal, we should then also argue why the conclusions we reach based on these special vector fields generalize—see [Lee12, Lem. 12.24].

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where the $U_i \in \mathfrak{X}(\mathcal{M})$ are arbitrary so long as $U_i(x) = u_i$.

Continuing with our examples, for a vector field $U \in \mathfrak{X}(\mathcal{M})$, the notation $u = U(x)$ normally refers to a tangent vector at x, while if we think of U as a tensor field, then $U(x)$ denotes the linear function $v \mapsto \langle u, v\rangle_x$ on the tangent space at x. The Riemannian metric is a tensor field of order two; let us call it G. Then, $G(x)$ is a bilinear function on $\mathrm{T}_x\mathcal{M} \times \mathrm{T}_x\mathcal{M}$ such that $G(x)(u, v) = \langle u, v\rangle_x$.

The map $x \mapsto T(x)$ is smooth, in a sense we now make precise. Similarly to Proposition 10.60, we can define a *tensor bundle* of order k over $\mathcal{M}$ as:

$$\begin{aligned} \mathrm{T}^k\mathrm{T}\mathcal{M} &= \left\{(x, L) : x \in \mathcal{M} \text{ and } L \in \mathrm{T}^k\mathrm{T}_x\mathcal{M}\right\}, \text{ where} \\ \mathrm{T}^k\mathrm{T}_x\mathcal{M} &= \left\{k\text{-linear functions from } (\mathrm{T}_x\mathcal{M})^k \text{ to } \mathbb{R}\right\}. \end{aligned} \tag{10.46}$$

Each tensor bundle can be endowed with a natural smooth manifold structure such that $\pi\colon \mathrm{T}^k\mathrm{T}\mathcal{M} \to \mathcal{M}$ defined by $\pi(x, L) = x$ is smooth. This is identical to how we equipped the tangent bundle $\mathrm{T}\mathcal{M}$ with such a smooth structure. Then, any section of $\mathrm{T}^k\mathrm{T}\mathcal{M}$, that is, any map T from $\mathcal{M}$ to $\mathrm{T}^k\mathrm{T}\mathcal{M}$ such that $\pi(T(x)) = x$ is called a *tensor field* of order k. This is commonly taken as the definition of a tensor field. *Smooth* tensor fields as we defined them above are exactly the smooth sections as defined here [Lee12, Prop. 12.19, Lem. 12.24].

By convention, $\mathrm{T}^0\mathrm{T}_x\mathcal{M} = \mathbb{R}$, so that $\mathrm{T}^0\mathrm{T}\mathcal{M} = \mathcal{M} \times \mathbb{R}$. Notice that $\mathrm{T}^1\mathrm{T}_x\mathcal{M}$ can be identified with $\mathrm{T}_x\mathcal{M}$, and that $\mathrm{T}^2\mathrm{T}_x\mathcal{M}$ can be identified with the set of linear maps from $\mathrm{T}_x\mathcal{M}$ into itself. Thus, $\mathrm{T}^1\mathrm{T}\mathcal{M}$ can be identified with $\mathrm{T}\mathcal{M}$ itself, and $\mathrm{T}^2\mathrm{T}\mathcal{M}$ can be identified as:

$$\mathrm{T}^2\mathrm{T}\mathcal{M} \equiv \{(x, L) : x \in \mathcal{M} \text{ and } L\colon \mathrm{T}_x\mathcal{M} \to \mathrm{T}_x\mathcal{M} \text{ is linear}\}. \tag{10.47}$$

Now that we think of smooth tensor fields as smooth maps on manifolds, it is natural to ask what happens if we differentiate them. In Chapter 5, we introduced the notion of connection ∇ on the tangent bundle $\mathrm{T}\mathcal{M}$. One can formalize the idea that ∇ induces a connection on any tensor bundle, unique once we require certain natural properties [Lee18, Prop. 4.15]. This gives meaning to the notation $\nabla_V T$ for $V \in \mathfrak{X}(\mathcal{M})$. Omitting quite a few details, we give an opportunistic construction of this object.

Recall how Vf is the derivative of a real function f against a vector field V. Since $T(U_1, \ldots, U_k)$ is a smooth function on $\mathcal{M}$, we can differentiate it against any smooth vector field V to obtain $VT(U_1, \ldots, U_k)$, also a smooth function on $\mathcal{M}$. The definition below is crafted to secure a natural chain rule for this differentiation.

Definition 10.77. *Given a smooth tensor field T of order k on a manifold $\mathcal{M}$ with a connection ∇, the* total covariant derivative ∇T *of T is a smooth tensor*

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field of order $k+1$ on $\mathcal{M}$, defined for all $U_1, \ldots, U_k, V \in \mathfrak{X}(\mathcal{M})$ by

$$\nabla T(U_1, \ldots, U_k, V) = VT(U_1, \ldots, U_k) - \sum_{i=1}^{k} T(U_1, \ldots, U_{i-1}, \nabla_V U_i, U_{i+1}, \ldots, U_k). \quad (10.48)$$

(If $\nabla T = 0$, we call T parallel.*) We also let $\nabla_V T$ be a smooth tensor field of order k (symmetric if T is symmetric), defined by*

$$(\nabla_V T)(U_1, \ldots, U_k) = \nabla T(U_1, \ldots, U_k, V).$$

(It is an exercise to check that these are indeed tensor fields.)

As an example, it is instructive to see how gradients and Hessians of scalar fields fit into the framework of covariant differentiation of tensor fields.

Example 10.78. *Let $\mathcal{M}$ be a Riemannian manifold with its Riemannian connection ∇. A smooth function $f\colon \mathcal{M} \to \mathbb{R}$ is a tensor field of order zero. Differentiating f as a tensor field using Definition 10.77, we find that ∇f is a tensor field of order one defined by*

$$\nabla f(U) = Uf = \mathrm{D}f(U) = \langle \operatorname{grad} f, U \rangle .$$

In other words, ∇f is the differential $\mathrm{D}f$, which we identify with the gradient vector field through the Riemannian metric. We now differentiate ∇f to produce $\nabla(\nabla f) = \nabla^2 f$: a tensor field of order two defined by

$$\begin{aligned} \nabla^2 f(U, V) &= V\nabla f(U) - \nabla f(\nabla_V U) \\ &= V\langle \operatorname{grad} f, U \rangle - \langle \operatorname{grad} f, \nabla_V U \rangle \\ &= \langle \nabla_V \operatorname{grad} f, U \rangle \\ &= \langle \operatorname{Hess} f(V), U \rangle . \end{aligned}$$

In other words, $\nabla^2 f$ and the Riemannian Hessian $\operatorname{Hess} f$ are identified through the Riemannian metric. This also shows that $\nabla^2 f$ is symmetric. Going one step further, we differentiate $\nabla^2 f$ to produce $\nabla^3 f = \nabla(\nabla^2 f)$: a tensor field of order three defined by

$$\begin{aligned} \nabla^3 f(U, V, W) &= W\nabla^2 f(U, V) \\ &\quad - \nabla^2 f(\nabla_W U, V) - \nabla^2 f(U, \nabla_W V) \\ &= W\langle \operatorname{Hess} f(V), U \rangle \\ &\quad - \langle \operatorname{Hess} f(V), \nabla_W U \rangle - \langle \operatorname{Hess} f(\nabla_W V), U \rangle \\ &= \langle \nabla_W(\operatorname{Hess} f(V)) - \operatorname{Hess} f(\nabla_W V), U \rangle . \end{aligned}$$

Notice that $\nabla^3 f$ is symmetric in its first two inputs, but not necessarily in its third input (see also Exercise 10.87). Based on the above, for a given $W \in \mathfrak{X}(\mathcal{M})$ it is useful to introduce $\nabla_W \operatorname{Hess} f\colon \mathfrak{X}(\mathcal{M}) \to \mathfrak{X}(\mathcal{M})$: an operator of the same type

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as the Riemannian Hessian itself given by

$$\nabla_W \mathrm{Hess} f(V) = \nabla_W(\mathrm{Hess} f(V)) - \mathrm{Hess} f(\nabla_W V). \tag{10.49}$$

The smooth tensor field $(U, V) \mapsto \langle \nabla_W \mathrm{Hess} f(V), U\rangle = \nabla^3 f(U, V, W)$ *is symmetric in* U *and* V*.*

Since tensor fields are pointwise objects, we can make sense of the notation $\nabla_v T$ for $v \in \mathrm{T}_x\mathcal{M}$ as follows: ∇T is a $(k+1)$-tensor field on $\mathcal{M}$, so that $(\nabla T)(x)$ is a $(k+1)$-linear map on $\mathrm{T}_x\mathcal{M}$; fixing the last input to be v, we are left with

$$(\nabla_v T)(u_1, \ldots, u_k) = \big((\nabla T)(x)\big)(u_1, \ldots u_k, v). \tag{10.50}$$

Thus, $\nabla_v T$ is a k-linear map on $\mathrm{T}_x\mathcal{M}$; if T is symmetric, so is $\nabla_v T$. In particular, $\nabla_v \mathrm{Hess} f$ is self-adjoint on $\mathrm{T}_x\mathcal{M}$ for $v \in \mathrm{T}_x\mathcal{M}$.

Given a curve c on $\mathcal{M}$, we defined the covariant derivative of a vector field along c in Section 5.7. This extends to tensors, in direct analogy with Theorem 5.29.

Definition 10.79. *Let* $c\colon I \to \mathcal{M}$ *be a smooth curve on a manifold* $\mathcal{M}$*. A* smooth tensor field Z of order k along c *is a map*

$$Z\colon \mathfrak{X}(c) \times \cdots \times \mathfrak{X}(c) \to \mathfrak{F}(I)$$

that is $\mathfrak{F}(I)$*-linear in each of its* k *inputs. We denote the set of such fields* $\mathfrak{X}^k(c)$*.*

Here too, we can reason that tensor fields are pointwise objects, in that $Z(t)$ is a k-linear map from $(\mathrm{T}_{c(t)}\mathcal{M})^k$ to $\mathbb{R}$: see [Lee18, Thm. 4.24, Prop. 5.15].

Theorem 10.80. *Let* $c\colon I \to \mathcal{M}$ *be a smooth curve on a manifold with a connection* ∇*. There exists a unique operator* $\frac{\mathrm{D}}{\mathrm{d}t}\colon \mathfrak{X}^k(c) \to \mathfrak{X}^k(c)$ *satisfying these properties for all* $Y, Z \in \mathfrak{X}^k(c)$*,* $T \in \mathfrak{X}^k(\mathcal{M})$*,* $g \in \mathfrak{F}(I)$ *and* $a, b \in \mathbb{R}$*:*

1. $\mathbb{R}$*-linearity:* $\frac{\mathrm{D}}{\mathrm{d}t}(aY + bZ) = a\frac{\mathrm{D}}{\mathrm{d}t}Y + b\frac{\mathrm{D}}{\mathrm{d}t}Z$*;*
2. *Leibniz rule:* $\frac{\mathrm{D}}{\mathrm{d}t}(gZ) = g'Z + g\frac{\mathrm{D}}{\mathrm{d}t}Z$*;*
3. *Chain rule:* $\big(\frac{\mathrm{D}}{\mathrm{d}t}(T \circ c)\big)(t) = \nabla_{c'(t)}T$ *for all* $t \in I$*.*

This operator is called the induced covariant derivative*.*

In the statement above, we understand $\nabla_{c'(t)}T$ through (10.50).

As a result of Definition 10.77, we also have the following chain rule: given

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$Z \in \mathfrak{X}^k(c)$ and $U_1, \ldots, U_k \in \mathfrak{X}(c)$:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\big(Z(U_1,\ldots,U_k)\big) &= \frac{\mathrm{d}}{\mathrm{d}t}\big(Z(t)(U_1(t),\ldots,U_k(t))\big) \\
&= \left(\frac{\mathrm{D}}{\mathrm{d}t}Z(t)\right)(U_1(t),\ldots,U_k(t)) \\
&\quad + Z(t)\left(\frac{\mathrm{D}}{\mathrm{d}t}U_1(t), U_2(t),\ldots,U_k(t)\right) + \cdots \\
&\quad + Z(t)\left(U_1(t),\ldots,U_{k-1}(t), \frac{\mathrm{D}}{\mathrm{d}t}U_k(t)\right) \\
&= \left(\frac{\mathrm{D}}{\mathrm{d}t}Z\right)(U_1,\ldots,U_k) \\
&\quad + Z\left(\frac{\mathrm{D}}{\mathrm{d}t}U_1, U_2,\ldots,U_k\right) + \cdots + Z\left(U_1,\ldots,U_{k-1}, \frac{\mathrm{D}}{\mathrm{d}t}U_k\right).
\end{aligned} \tag{10.51}$$

Example 10.81. *If $c\colon I \to \mathcal{M}$ is a geodesic on the Riemannian manifold $\mathcal{M}$ with Riemannian connection ∇ and $f\colon \mathcal{M} \to \mathbb{R}$ is smooth, we have:*

$$\begin{aligned}
(f\circ c)' &= \nabla_{c'} f = (\nabla f \circ c)(c'), \\
(f\circ c)'' &= (\nabla_{c'}\nabla f)(c') + (\nabla f \circ c)(c'') = (\nabla^2 f \circ c)(c', c'), \\
(f\circ c)''' &= (\nabla_{c'}\nabla^2 f)(c', c') + (\nabla^2 f \circ c)(c'', c') + (\nabla^2 f \circ c)(c', c'') \\
&= (\nabla^3 f \circ c)(c', c', c').
\end{aligned}$$

In particular, if $c(0) = x$ and $c'(0) = u$, then at $t = 0$ it follows that:

$$\begin{aligned}
(f\circ c)'(0) &= \nabla f(x)(u) = \langle \mathrm{grad} f(x), u\rangle_x, \\
(f\circ c)''(0) &= \nabla^2 f(x)(u,u) = \langle \mathrm{Hess} f(x)[u], u\rangle_x, \\
(f\circ c)'''(0) &= \nabla^3 f(x)(u,u,u) = \langle (\nabla_u \mathrm{Hess} f)[u], u\rangle_x,
\end{aligned}$$

with $\nabla_u \mathrm{Hess} f$ as defined through (10.49) and (10.50).

Circling back to Section 10.4, let us now discuss Lipschitz continuity of tensor fields. As afforded by Remark 8.6, here we do not require tensor fields to be smooth (that is, infinitely differentiable). We understand the differentiability properties of tensor fields as maps between manifolds, as outlined around (10.46). We start with a definition.

Definition 10.82. *A tensor field T of order k on a Riemannian manifold $\mathcal{M}$ with its Riemannian connection is* L-Lipschitz continuous *if for all (x, s) in the domain of the exponential map and for all $u_1, \ldots, u_k \in \mathrm{T}_x\mathcal{M}$ we have*

$$|T(\mathrm{Exp}_x(s))(P_s u_1, \ldots, P_s u_k) - T(x)(u_1,\ldots,u_k)| \leq L\|s\|_x \|u_1\|_x \cdots \|u_k\|_x,$$

where P_s is parallel transport along $\gamma(t) = \mathrm{Exp}_x(ts)$ from $t = 0$ to $t = 1$.

It is an exercise to show that this definition is compatible with the ones we introduced earlier, for example for f, $\mathrm{grad} f$ and $\mathrm{Hess} f$.

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Assume T is differentiable. Let c be a smooth curve on $\mathcal{M}$ satisfying $c(0) = x$ and $c'(0) = s$. Parallel transport $u_1, \ldots, u_k \in \mathrm{T}_x\mathcal{M}$ along c to form $U_i(t) = \mathrm{PT}^c_{t\leftarrow 0} u_i$ for $i = 1, \ldots, k$. Then, owing to $\frac{\mathrm{D}}{\mathrm{d}t} U_i = 0$ for all i we have (using (10.51) and the chain rule in Theorem 10.80):

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\Big((T \circ c)(U_1, \ldots, U_k)\Big) &= \left(\frac{\mathrm{D}}{\mathrm{d}t}(T \circ c)\right)(U_1, \ldots, U_k) \\ &= (\nabla_{c'} T)(U_1, \ldots, U_k) \\ &= (\nabla T \circ c)(U_1, \ldots, U_k, c'). \end{aligned} \tag{10.52}$$

In particular, at $t = 0$ this says:

$$\nabla T(x)(u_1, \ldots, u_k, s) = (\nabla_s T)(u_1, \ldots, u_k) = \frac{\mathrm{d}}{\mathrm{d}t}(T \circ c)(U_1, \ldots, U_k)\Big|_{t=0}. \tag{10.53}$$

In light of these identities, we have the following simple observations.

Proposition 10.83. *Let T be a tensor field of order k on $\mathcal{M}$ (Riemannian).*

1. *If T is differentiable and L-Lipschitz continuous, then ∇T is bounded by L, that is, for all $x \in \mathcal{M}$ and for all $u_1, \ldots u_k, s \in \mathrm{T}_x\mathcal{M}$,*

$$|\nabla T(x)(u_1, \ldots, u_k, s)| \;=\; |(\nabla_s T)(u_1, \ldots, u_k)| \;\leq\; L\|s\|_x \|u_1\|_x \cdots \|u_k\|_x.$$

2. *If T is continuously differentiable and ∇T is bounded by L, then T is L-Lipschitz continuous.*

Proof. For the first claim, use (10.53), continuity of the absolute value function and Definition 10.82 with $c(t) = \mathrm{Exp}_x(ts)$ to see that

$$\begin{aligned}|\nabla T(x)(u_1, \ldots, u_k, s)| &= |(\nabla_s T)(u_1, \ldots, u_k)| \\ &= \lim_{t \to 0} \frac{|T(\mathrm{Exp}_x(ts))(P_{ts}u_1, \ldots, P_{ts}u_k) - T(x)(u_1, \ldots, u_k)|}{|t|} \\ &\leq \lim_{t \to 0} \frac{1}{|t|} L\|ts\|_x \|u_1\|_x \cdots \|u_k\|_x = L\|s\|_x \|u_1\|_x \cdots \|u_k\|_x,\end{aligned}$$

as announced.

For the second claim, a bit of standard calculus and (10.52) (including notation there) yield

$$\begin{aligned}&T(c(1))(U_1(1), \ldots, U_k(1)) - T(x)(u_1, \ldots, u_k) \\ &\quad = \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\Big(T(c(t))(U_1(t), \ldots, U_k(t))\Big)\mathrm{d}t \\ &\quad = \int_0^1 (\nabla T \circ c)(U_1, \ldots, U_k, c')(t)\mathrm{d}t.\end{aligned}$$

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Therefore, using that ∇T is bounded by L,

$$\begin{aligned}
&|T(c(1))(U_1(1), \dots, U_k(1)) - T(x)(u_1, \dots, u_k)| \\
&\quad \leq \int_0^1 L\|U_1(t)\|_{c(t)} \cdots \|U_k(t)\|_{c(t)} \|c'(t)\|_{c(t)} \mathrm{d}t \\
&\quad = L\|u_1\|_x \cdots \|u_k\|_x \cdot L(c),
\end{aligned}$$

where $L(c)$ is the length of the curve c over $[0, 1]$. The claim follows in particular by letting $c(t) = \mathrm{Exp}_x(ts)$, in which case $L(c) = \|s\|_x$. □

Remark 10.84. *The Riemannian metric establishes a one-to-one correspondence between the tangent vector $u \in \mathrm{T}_x\mathcal{M}$ and the linear map $v \mapsto \langle u, v\rangle_x$ from $\mathrm{T}_x\mathcal{M}$ to $\mathbb{R}$. In the absence of a metric, we distinguish between these two types of objects, respectively called vectors and* covectors. *When doing so, it is useful to define tensor fields as maps that transform vector fields and/or covector fields into scalar fields, leading to the notions of* covariant, contravariant and mixed tensor fields. *These terms are likely to come up in discussions of tensor fields on Riemannian manifolds as well, because they are often familiar to readers from non-Riemannian smooth geometry. See [Lee12, Ch. 12] and [Lee18, Ch. 4, Ch. 5] for details.*

Exercise 10.85. *Check that ∇T as provided by Definition 10.77 is indeed a tensor field of order $k+1$, and that if T is symmetric, then $\nabla_V T$ is a symmetric tensor field of order k.*

Exercise 10.86. *Let $\mathcal{M}$ be a Riemannian manifold and let G be the smooth tensor field of order two defined by $G(U, V) = \langle U, V\rangle$. Check that a connection ∇ on $\mathcal{M}$ is compatible with the metric if and only if G is parallel. (See also [Lee18, Prop. 5.5] for further characterizations of compatibility between the metric and the connection.)*

Exercise 10.87. *Let $f : \mathcal{M} \to \mathbb{R}$ be three times continuously differentiable on a Riemannian manifold $\mathcal{M}$ equipped with a retraction* R. *Assume*

$$\left| f(\mathrm{R}_x(s)) - f(x) - \langle s, \mathrm{grad} f(x)\rangle_x - \frac{1}{2}\langle s, \mathrm{Hess} f(x)[s]\rangle_x \right| \leq \frac{L}{6}\|s\|_x^3 \tag{10.54}$$

for all (x, s) in a neighborhood of the zero section in the tangent bundle. (Compare with regularity assumption A6.7 on p135; see also Exercise 10.58.) Further assume R *is a* third-order retraction, *which we define to be a second-order retraction for which all curves $c(t) = \mathrm{R}_x(ts)$ with $(x, s) \in \mathrm{T}\mathcal{M}$ obey $c'''(0) = 0$, where $c''' = \frac{\mathrm{D}}{\mathrm{d}t}c''$. In particular, the exponential map is a third-order retraction (see also Exercise 10.88). Show for all $(x, s) \in \mathrm{T}\mathcal{M}$ that*

$$|\langle(\nabla_s \mathrm{Hess} f)[s], s\rangle_x| = |\nabla^3 f(x)(s, s, s)| \leq L\|s\|_x^3. \tag{10.55}$$

In other words: the symmetric part of $\nabla^3 f(x)$ *is bounded by L.*

Conversely, show that if (10.55) holds for all $(x, s) \in \mathrm{T}\mathcal{M}$, then (10.54) holds with $\mathrm{R} = \mathrm{Exp}$ *for all (x, s) in the domain of* Exp.

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We state here without details that $\nabla^3 f(x)$ may not be symmetric when x is not a critical point of f and also when $\mathcal{M}$ is not flat at x (this follows from the Ricci identity *applied to ∇f). Therefore, $\nabla^3 f(x)$ may not be bounded by L even if its symmetric part is. Consequently, the developments here do not allow us to conclude as to the Lipschitz properties of* Hess$f$ *(which would have otherwise followed from Corollary 10.52 or Proposition 10.83).*

Exercise 10.88. *Consider a Riemannian submanifold $\mathcal{M}$ of a Euclidean space $\mathcal{E}$. Given $(x, v) \in \mathrm{T}\mathcal{M}$, let $c\colon I \to \mathcal{M}$ be a smooth curve on $\mathcal{M}$ such that $c(0) = x$ and $x + tv - c(t)$ is orthogonal to $\mathrm{T}_{c(t)}\mathcal{M}$ for all t: this is the case if $c(t)$ is a curve obtained through metric projection retraction (see Section 5.12). Show that $c'''(0) = -2\mathcal{W}\big(v, \frac{\mathrm{d}}{\mathrm{d}t}c'(0)\big)$, where $\mathcal{W}$ is the Weingarten map (5.38). In general, this is nonzero. Thus, we do not expect the metric projection retraction to be third order in general. Hint: use (5.18) to express $c'''(0)$ in terms of extrinsic derivatives of c, and simplify that expression by computing one more derivative of g in the proof of Proposition 5.55.*

Exercise 10.89. *Given a vector field V on $\mathcal{M}$ (Riemannian), let $T = \langle V, \cdot \rangle$ be the associated tensor field of order one. Show that T is L-Lipschitz continuous in the sense of Definition 10.82 if and only if V is L-Lipschitz continuous in the sense of Definition 10.44. Likewise, for each $x \in \mathcal{M}$, let $H(x)$ denote a linear map from $\mathrm{T}_x\mathcal{M}$ into itself, and let T denote the associated tensor field of order two defined through $T(x)(u, v) = \langle H(x)(u), v \rangle_x$. Show that T is L-Lipschitz continuous in the sense of Definition 10.82 if and only if H is L-Lipschitz continuous in the sense of Definition 10.49.*

10.8 Notes and references

Considering geodesics as length-minimizing curves, it is possible to generalize the concept of geodesic to arbitrary metric spaces, specifically, without the need for a smooth or Riemannian structure. See for example the monograph by Bacák [Bac14] for an introduction to *geodesic metric spaces*, and applications in convex analysis and optimization on Hadamard spaces.

For a connected manifold, there always exists a Riemannian metric which makes it complete [NO61].

Propositions 10.23 and 10.26 (and their proofs and corollaries) were designed with Eitan Levin and also discussed with Stephen McKeown. They are inspired by the proof of the Tubular Neighborhood Theorem in [Lee18, Thm. 5.25] and [Pet06, Prop. 5.18]. They notably imply that the injectivity radius function $\mathrm{inj}\colon \mathcal{M} \to \mathbb{R}$ is lower-bounded by a positive, continuous function, with (one could argue) fewer technicalities than are required to prove inj itself is continuous.

Many Riemannian geometry textbooks restrict their discussion of the injectivity radius to connected and complete manifolds. For disconnected manifolds, we defined *complete* to mean geodesically complete, i.e., each connected component

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is metrically complete. A function is continuous on a disconnected set if it is continuous on each connected component. Thus, continuity of inj on connected and complete manifolds implies continuity on complete manifolds. It is easy to find a published proof that the function $\mathrm{inj}\colon \mathcal{M} \to \mathbb{R}$ is continuous if $\mathcal{M}$ is (connected and) complete (see for example [Lee18, Prop. 10.37]), but it has proven difficult to locate one that applies to incomplete manifolds as well. The claim appears without proof in [Cha06, Thm. III.2.3]. Following a discussion, Stephen McKeown provided[8] a proof that inj is lower-semicontinuous, then John M. Lee added a proof that inj is upper-semicontinuous, together confirming that it is continuous: see Lemmas 10.90 and 10.91 below. Both proofs rely on continuity in the complete case. They are rearranged to highlight commonalities.

Our main motivation to study continuity of inj is to reach Corollary 10.25, stating that the map $(x, y) \mapsto \mathrm{Log}_x(y)$ is smooth over the specified domain. O'Neill makes a similar statement: pick an *open* set $S \subseteq \mathcal{M}$; that set is deemed *convex* (by [O'N83, Def. 5.5]) if, for all $x \in S$, the map Exp_x is a diffeomorphism from some neighborhood of the origin in $\mathrm{T}_x\mathcal{M}$ to S. Then, $(x, v) \mapsto (x, \mathrm{Exp}_x(v))$ is a diffeomorphism from the appropriate set in $\mathrm{T}S$ to $S \times S$ [O'N83, Lem. 5.9]. This shows the map $L\colon S \times S \to \mathrm{T}S$ such that $L(x, y) \in \mathrm{T}_x\mathcal{M}$ is the initial velocity of the (unique) geodesic $\gamma\colon [0, 1] \to S$ connecting x to y is smooth: L is also a kind of inverse for the exponential (though not necessarily the same as Log).

The tool of choice to differentiate the exponential map (and the logarithmic map) is Jacobi fields [Lee18, Prop. 10.10]. Some examples of this are worked out in the context of optimization on manifolds in [CB22a] and [LC20].

Parallel transporting a tangent vector u at x to all the points in a normal neighborhood [Lee18, p131] of x along geodesics through x contained in that neighborhood results in a smooth vector field. This is a well-known fact; details of the argument appear notably in [LB20, Lem. A.1].

The Riemannian notion of Lipschitz continuous gradient (and also Lipschitz continuous vector field) appears in [dCN95, Def. 3.1, p79] and [dCNdLO98, Def. 4.1], with a definition equivalent to the characterization we give in Proposition 10.45. This may be their first occurrence in an optimization context. There too, the motivation is to derive inequalities such as the ones in Proposition 10.53. Such inequalities appear often in optimization papers, see also [AMS08, §7.4], [SFF19, App. A] and [ABBC20], among many others. Lipschitz continuous Hessians appear in an optimization context in [FS02, Def. 2.2], in line with our characterization in Proposition 10.50. A general definition of Lipschitz continuous maps in tangent bundles of any order (covering tensors fields of any order) appears in [RW12], in the preliminaries on geometry. A general notion of 'fundamental theorem of calculus' for tensor fields on Riemannian manifolds, based on parallel transport, is spelled out in [ABM08, eq. (2.3)].

As an alternative to Definition 10.44, one could also endow the tangent bundle

[8] mathoverflow.net/questions/335032

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with a metric space structure, so that we can then apply the standard notion of Lipschitz continuity to vector fields as maps between two metric spaces. A canonical choice would follow from the *Sasaki metric* [GHL04, §2.B.6]. See [dOF20] for a comparison of the two concepts.

The notion of vector transport appears in [AMS08, §8]. The related notion of transporter is introduced in [QGA10a] with reference to a linear structure space, and further developed in [HGA15] for general manifolds. The constructions of transporters from other transporters via inversions and adjoints are natural extensions.

Corollary 10.75 handles finite differences of the Riemannian Hessian using an arbitrary transporter (Definition 10.61). An analogous result for vector transports (Definition 10.68) appears in [AMS08, Lem. 8.2.2].

In Exercise 10.73, we contemplate the role of the initial acceleration of curves of the form $c(t) = \mathrm{R}_x(s+t\dot{s})$. Consider the special case where R is the exponential map. If $s = 0$, then c is a geodesic so that $c''(0) = 0$; but for $s \neq 0$ we expect $c''(0) \neq 0$ in general. The tangent vector $c''(0)$ and its norm are tightly related to curvature of the manifold. See [CB22a, LC20] for a discussion of that vector and its effects on the Lipschitzness of pullbacks and their derivatives.

In Section 10.7, we follow do Carmo [dC92, §4.5], in that we rely on the Riemannian metric to avoid the need to distinguish between vectors and covectors, and we build up the differentiation of tensor fields by quoting the desired chain rule directly, bypassing many technical steps. This simplifies the discussion without loss of generality.

We close this section with the proofs of continuity of the injectivity radius, as stated in Proposition 10.24. In these proofs, we do not need to worry about infinite values. Indeed, if $\mathrm{inj}(x) = \infty$ at some point x, then Exp_x is defined on all of $\mathrm{T}_x\mathcal{M}$. Thus, the connected component of x is complete [Lee18, Cor. 6.20], and it follows by [Lee18, Prop. 10.37] that inj is continuous on that component. (More specifically: inj is infinite at all points in that component.)

Lemma 10.90. *The injectivity radius function* $\mathrm{inj}\colon \mathcal{M} \to (0, \infty]$ *is lower-semi-continuous.*

Proof by Stephen McKeown. For contradiction, assume inj is not lower-semi-continuous at some point $x \in \mathcal{M}$. Then, there exists a sequence of points $x_0, x_1, x_2, \ldots$ on $\mathcal{M}$ such that

$$\lim_{k\to\infty} x_k = x \qquad \text{and} \qquad \forall k,\ \mathrm{inj}(x_k) \leq r < R = \mathrm{inj}(x).$$

Define $\varepsilon = \frac{R-r}{3}$ and $r' = r+\varepsilon$, $r'' = r+2\varepsilon$ so that $r < r' < r'' < R$. By definition of R, the exponential map Exp_x induces a diffeomorphism $\varphi\colon B(x, R) \to B^d(R)$: from the open geodesic ball on $\mathcal{M}$ centered at x with radius R to the open Euclidean ball in $\mathbb{R}^d$ centered at the origin with radius R, where $d = \dim \mathcal{M}$. (ϕ is constructed from the inverse of Exp_x followed by a linear isometry from $\mathrm{T}_x\mathcal{M}$ to $\mathbb{R}^d$.) Let g denote the Riemannian metric on $\mathcal{M}$, let $\tilde{g}$ denote the pushforward

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of $g|_{B(x,R)}$ to $B^d(R)$ (through φ), and let g_0 denote the Euclidean metric on $\mathbb{R}^d$. Consider a smooth bump function $\chi\colon \mathbb{R}^d \to [0,1]$ whose value is 1 on the closed ball $\bar{B}^d(r'')$ and with support in $\bar{B}^d(R)$ [Lee12, Prop. 2.25]. Then,

$$\hat{g} = \chi\tilde{g} + (1-\chi)g_0$$

is a Riemannian metric on $\mathbb{R}^d$ such that $\hat{\mathcal{M}} = (\mathbb{R}^d, \hat{g})$ is a (connected) complete Riemannian manifold [Lee18, Pb. 6-10].

Consequently, the injectivity radius function $\hat{\text{inj}}\colon \hat{\mathcal{M}} \to (0,\infty]$ is continuous [Lee18, Prop. 10.37]. Furthermore, since the metrics g and $\hat{g}$ agree (through φ) on $\bar{B}(x,r'')$ and $\bar{B}^d(r'')$, we deduce that for all $y \in \mathcal{M}$ and $\rho > 0$ it holds that

$$B(y,\rho) \subseteq B(x,r'') \qquad \Longrightarrow \qquad \begin{cases} \hat{\text{inj}}(\hat{y}) = \text{inj}(y) & \text{if } \text{inj}(y) < \rho, \\ \hat{\text{inj}}(\hat{y}) \geq \rho & \text{otherwise,} \end{cases}$$

where $\hat{y} \in \mathbb{R}^d$ is the image of y through φ.

We use this fact in two ways:

1. $B(x,r'') \subseteq B(x,r'')$ and $\text{inj}(x) = R > r''$, hence $\hat{\text{inj}}(\hat{x}) \geq r''$, and
2. There exists k_0 large enough such that, for all $k \geq k_0$, $\text{dist}(x_k, x) < \varepsilon$, so that $B(x_k,r') \subset B(x,r'')$. Moreover, $\text{inj}(x_k) \leq r < r'$, so that $\hat{\text{inj}}(\hat{x}_k) = \text{inj}(x_k) \leq r$ for all $k \geq k_0$.

Together with the fact that $\hat{\text{inj}}$ is continuous, these yield:

$$r < r'' \leq \hat{\text{inj}}(\hat{x}) = \lim_{k\to\infty} \hat{\text{inj}}(\hat{x}_k) \leq r,$$

a contradiction. □

Lemma 10.91. *The injectivity radius function* $\text{inj}\colon \mathcal{M} \to (0,\infty]$ *is upper-semicontinuous.*

Proof by John M. Lee. The proof parallels that of lower-semicontinuity. For contradiction, assume inj is not upper-semicontinuous at some point $x \in \mathcal{M}$. Then, there exists a sequence of points $x_0, x_1, x_2, \ldots$ on $\mathcal{M}$ such that

$$\lim_{k\to\infty} x_k = x \qquad \text{and} \qquad \forall k,\ \text{inj}(x_k) \geq R > r = \text{inj}(x).$$

Define $\varepsilon = \frac{R-r}{3}$ and $r' = r + \varepsilon$, $r'' = r + 2\varepsilon$ so that $r < r' < r'' < R$. Because the injectivity radius at x is smaller than at the points in the sequence, it is not enough to consider Exp_x to setup a diffeomorphism with a ball in $\mathbb{R}^d$. Instead, we pick a special point in the sequence to act as a center. Let k_0 be large enough so that $\text{dist}(x_k, x) < \varepsilon/2 < \varepsilon$ for all $k \geq k_0$. By triangular inequality, we also have $\text{dist}(x_k, x_{k_0}) < \varepsilon$ for all $k \geq k_0$. Now, use $\text{Exp}_{x_{k_0}}$ to setup a diffeomorphism $\varphi\colon B(x_{k_0}, R) \to B^d(R)$: we can do this since $\text{inj}(x_{k_0}) \geq R$. Let g denote the Riemannian metric on $\mathcal{M}$, let $\tilde{g}$ denote the pushforward of $g|_{B(x_{k_0},R)}$ to $B^d(R)$ (through φ), and let g_0 denote the Euclidean metric on $\mathbb{R}^d$. Consider a smooth

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bump function $\chi\colon \mathbb{R}^d \to [0,1]$ whose value is 1 on the closed ball $\bar{B}^d(r'')$ and with support in $\bar{B}^d(R)$. Then,

$$\hat{g} = \chi\tilde{g} + (1-\chi)g_0$$

is a Riemannian metric on $\mathbb{R}^d$ such that $\hat{\mathcal{M}} = (\mathbb{R}^d, \hat{g})$ is a (connected) complete Riemannian manifold.

Consequently, the injectivity radius function $\hat{\text{inj}}\colon \hat{\mathcal{M}} \to (0,\infty]$ is continuous. Furthermore, since the metrics g and $\hat{g}$ agree (through φ) on $\bar{B}(x_{k_0}, r'')$ and $\bar{B}^d(r'')$, we deduce that for all $y \in \mathcal{M}$ and $\rho > 0$ it holds that

$$B(y,\rho) \subseteq B(x_{k_0}, r'') \qquad \Longrightarrow \qquad \begin{cases} \hat{\text{inj}}(\hat{y}) = \text{inj}(y) & \text{if } \text{inj}(y) < \rho, \\ \hat{\text{inj}}(\hat{y}) \geq \rho & \text{otherwise,} \end{cases}$$

where $\hat{y} \in \mathbb{R}^d$ is the image of y through φ.

We use this fact in two ways:

1. $B(x,r') \subset B(x_{k_0}, r'')$ and $\text{inj}(x) = r < r'$, hence $\hat{\text{inj}}(\hat{x}) = \text{inj}(x) = r$, and
2. For all $k \geq k_0$, $B(x_k, r') \subset B(x_{k_0}, r'')$ and $\text{inj}(x_k) \geq R > r'$, thus $\hat{\text{inj}}(\hat{x}_k) \geq r'$.

Together with the fact that $\hat{\text{inj}}$ is continuous, these yield:

$$r = \text{inj}(x) = \hat{\text{inj}}(\hat{x}) = \lim_{k\to\infty} \hat{\text{inj}}(\hat{x}_k) \geq r' > r,$$

a contradiction. □